

# Voters Know the Difference: National Shocks and Local Accountability in Brazil

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**Abstract:** This paper examines whether local exposure to Brazil’s 2014–2016 recession affected incumbent mayors’ electoral outcomes in the 2016 municipal elections. The recession was driven by the end of the commodities boom and *Lava Jato*, Brazil’s landmark anti-corruption investigation. Using shift-share measures of local exposure to the *Lava Jato*-driven recession and China trade shocks, I find that national economic shocks did not significantly affect incumbent mayors’ vote shares, but did influence the 2018 presidential election. These results are consistent with voters exercising sophistication in attributing responsibility across levels of government. Furthermore, political information amplified economic voting. Mayors from parties with no *Lava Jato* defendants received an electoral premium, and there is suggestive evidence that left-wing *Lava Jato* parties faced greater punishment than their center or right-wing counterparts. In municipalities where mass protests against corruption were more widespread, voters in more recession-exposed areas punished incumbents significantly more than those in less exposed ones.

**JEL Codes:** D72, H70, F16, D73

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# 1 Introduction

When a national recession occurs, the literature on economic voting suggests that voters punish the president (Healy and Malhotra, 2013), but a small literature has examined whether such shocks also affect local politicians, such as mayors (De Benedictis-Kessner and Warshaw, 2020). To study this effect, I use Brazil’s multifaceted recession between 2014 and 2016 as a case study for two main reasons. First, the downturn was driven by two main shocks: the end of the commodities supercycle<sup>1</sup>, and the largest Brazilian anti-corruption probe *Lava Jato* that amplified economic uncertainty. Second, the recession coincides with the 2016 municipal elections, providing a clear source of exogenous identification.

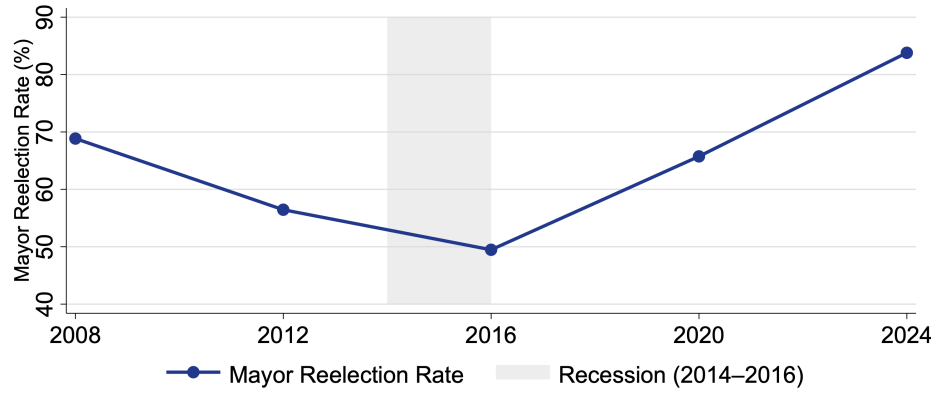
In the 2016 municipal elections, the mayor’s reelection rate and the change in the incumbent mayor’s vote share were both at their lowest across the last five election cycles (Figure 1). Conditional on running, only 50% of mayors won re-election, and incumbent mayors’ vote share declined by 7.2 percentage points relative to the previous election on average. This suggests that voters may have punished mayors for a concurrent national recession.

This paper tests whether national economic shocks influenced local electoral accountability in the 2016 municipal elections, and whether information about corruption during *Lava Jato* investigations amplified economic voting at the local level. To exploit these exogenous shocks, I construct two measures of local shock exposure. First, I employ a Bartik shift-share (Bartik, 1991) to compute local exposure to the *Lava Jato*-driven recession across Brazilian municipalities. Second, following the China shock literature (e.g. Autor et al., 2013; Costa et al., 2016; Autor et al., 2020), I construct an import supply shock and an export demand shock from China. To isolate variation originating from China, I implement an instrumental variable strategy that exploits variation in China’s bilateral trade with all countries excluding Brazil, sweeping out world-level shocks and Brazilian demand and supply endogeneity. The two shocks capture distinct sectoral distribu-

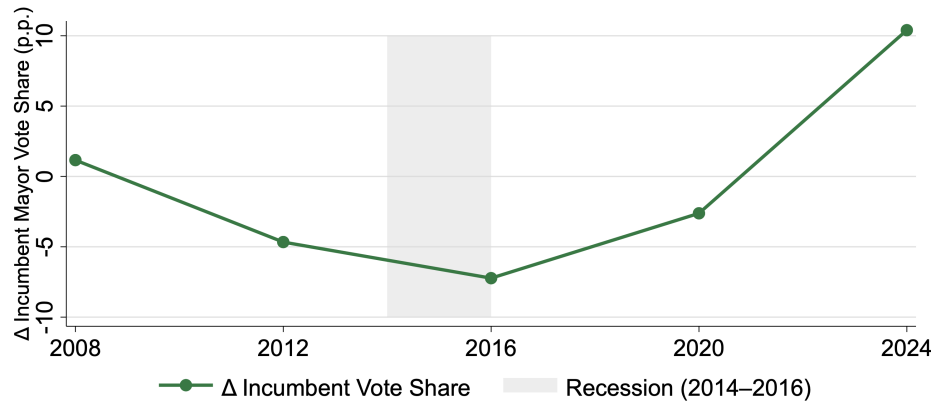
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<sup>1</sup>China’s rapid industrialisation in the early 2000s drove up global commodity demand and prices. Brazil, as a major commodities exporter to China, benefited substantially from this boom. However, as Chinese growth softened in the early 2010s, the cycle ended.

Figure 1: Local Electoral Accountability and the 2014–2016 Recession



(a) Mayor Reelection Rate



(b)  $\Delta$  Incumbent Mayor Vote Share

*Notes:* Panel (a) shows the mayor reelection rate across municipal election cycles, conditional on the incumbent mayor running. Panel (b) shows the average change in the incumbent mayor's vote share between consecutive elections, conditional on the incumbent mayor running. The shaded area denotes the 2014–2016 recession period as defined by [Trece et al. \(2024\)](#). Data on electoral outcomes are from TSE.

tions of exposure. Recession exposure is most sensitive to the construction and oil & gas sectors, which were among the most heavily investigated by *Lava Jato*. Meanwhile, the Chinese import supply shock is concentrated in manufacturing industries, and the Chinese export demand shock is concentrated in agriculture.

I find that *Lava Jato*-driven recession exposure did not affect the change in the incumbent mayor's vote share on average, either in the main specification or across robustness checks. Furthermore, although China trade shocks significantly affected local labour markets, they did not influence municipal election outcomes in 2016. Neither shock produced statistically significant effects when the incumbent belonged to *Partido dos Trabalhadores* (PT), the left-wing presidential party. Nevertheless, *Lava Jato*-driven recession exposure significantly shaped the subsequent presidential election in 2018: municipalities at the 75th percentile punished PT by between 2.60 and 3.21 percentage points more than those at the 25th percentile. Similarly, import supply shocks punished the president while export demand shocks rewarded the president. Overall, these findings provide consistent evidence that voters exercise a degree of sophistication in attributing responsibility across levels of government, holding the president rather than local mayors accountable for national economic shocks.

Since the largest anti-corruption investigation in Brazilian history was unfolding during that period ([Valarini and Pohlmann, 2019](#)), I test whether political information amplified mayoral punishment for the recession. Municipalities in the top 25% of cell-phone and computer penetration present a more negative coefficient of recession exposure on incumbent vote share, consistent with greater punishment, though the estimates are not statistically significant. While general information access does not tell a clear story, information specific to corruption yields a more convincing mechanism. Mayors from parties with no *Lava Jato* defendants received an electoral premium: municipalities at the 75th percentile of recession exposure rewarded incumbents by 9.13 percentage points more than those at the 25th percentile. However, voters did not systematically punish mayors belonging to *Lava Jato*-investigated parties. One possible explanation is that party labels are a weak signal of federal politics at the local level, given the high rate of party switching among



mayors. Nonetheless, there is suggestive evidence that voters punished left-wing *Lava Jato* parties more than their center or right-wing counterparts, though the difference is not statistically significant. This pattern is consistent with the rise of right-wing populism following *Lava Jato* argued by [Barros and Santos Silva \(2025\)](#). Furthermore, amid rising media coverage of corruption, mass protests took place between 2015 and 2017 against the PT federal government and in support of *Lava Jato* investigations. In municipalities where more than 7% of the population attended protests — approximately the top 10% of the distribution — those at the 75th percentile of recession exposure punished incumbents by 22.71 percentage points more than those at the 25th percentile. These findings suggest that mayoral punishment for the recession is amplified in municipalities where information about corruption is more prevalent.

This study contributes to the literature in three ways. First, methodologically, I apply the [Bartik \(1991\)](#) shift-share design to retrospective voting in local elections, and I am the first to implement the decomposed Chinese trade shocks of [Costa et al. \(2016\)](#) to study political outcomes in a developing country, to the best of my knowledge. Second, despite a growing literature examining the effects of recession and trade liberalization on Brazilian presidential elections ([Costa et al., 2023](#); [Ogeda et al., 2024](#); [Barros and Santos Silva, 2025](#)), this is the first paper to study the impact of national economic shocks on municipal elections in Brazil. Third, I contribute to the literature on the information channel of economic voting ([De Benedictis-Kessner and Warshaw, 2020](#)) by examining how the *Lava Jato* investigation shaped local electoral accountability.

The remainder of the paper is organized as follows. Section 2 provides the institutional background and literature review. Section 3 describes the electoral, employment, and trade data. Section 4 presents the *Lava Jato*-driven recession exposure and China shock estimation strategies. Section 5 reports the main results for both strategies on municipal and presidential elections. Section 6 discusses the information mechanism. Section 7 concludes. Appendix A provides additional figures and results.

## 2 Background

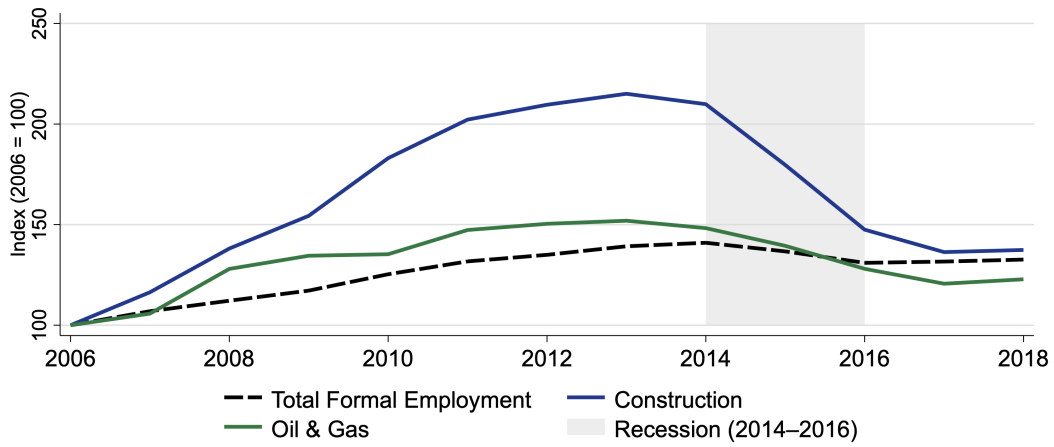
### 2.1 Institutional Background

Brazil experienced sustained employment growth during the 2000s, largely driven by the commodities boom (Barros and Santos Silva, 2025). As China underwent rapid industrialization in the early 2000s, demand for commodities rose, driving up global prices. Brazil, as a major commodities exporter to China, benefited substantially from this boom. Figure 2 shows a 39 percentage point increase in total formal employment from 2006 to 2013. Amid this growth, rising social spending and the implementation of *Bolsa Família*, the world’s largest cash-transfer program, reduced poverty and inequality (e.g. Ferreira et al., 2021; Best et al., 2026).

Yet, the economy underwent a national recession between the first quarter of 2014 and the fourth quarter of 2016 (Trece et al., 2024). Figure 2 shows a 6% decline in formal employment. The literature attributes the recession to a combination of three factors (Barros and Santos Silva, 2025). First, macroeconomic mismanagement, including the federal government’s violation of the fiscal responsibility law. Second, as Chinese growth softened in the early 2010s, the commodities supercycle ended. Figure 3 indicates that Brazil’s imports from China decreased by 38% and exports to China by 24% during the recession period.

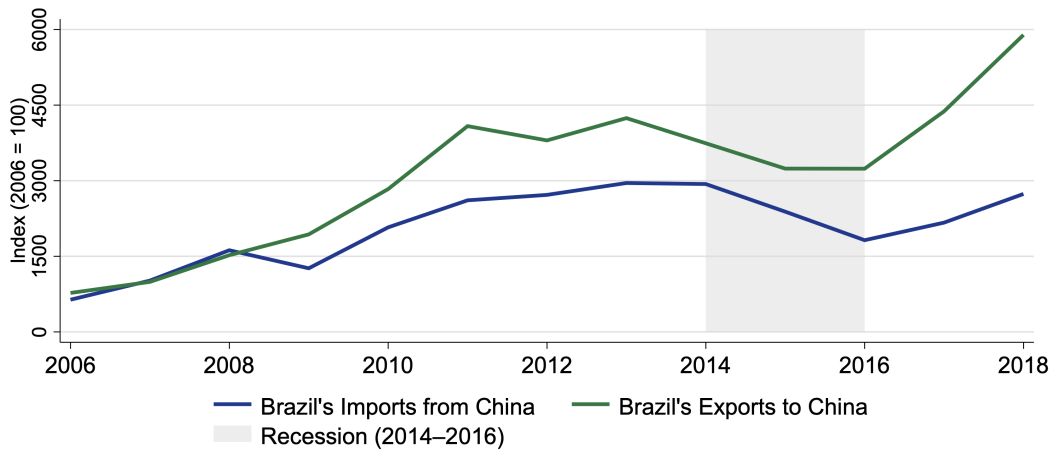
Third, *Lava Jato* was the largest anti-corruption and anti-money laundering investigation in Brazil, and arguably the most complex in Latin America. It began in 2014 when the Federal Police and *Ministério Público Federal* investigated an illegal money laundering scheme operated through black market dealers at a chain of gas stations. Those dealers’ bank accounts had frequent transactions with the largest Brazilian construction companies and the state-owned oil company Petrobras. The dealers subsequently signed a plea bargain that revealed the full scale of corruption involving politicians, political parties, state governors, Congress and the federal government (Valarini and Pohlmann, 2019; Ferraz et al., 2025). Following this large scandal, widespread political and economic uncertainty ensued. Firms under investigation lost access to government contracts and experienced declines in employment and reduced access to credit (Ferraz et al., 2025). Employ-

Figure 2: Formal Employment Change



*Notes:* The figure plots an index of formal employment (2006 = 100) by sector between 2006 and 2018. Total formal employment, construction, and oil & gas sectors are shown separately. Oil & gas is defined by CNAE 2-digit codes 06 (extraction of crude petroleum and natural gas), 09 (mining support service activities), and 19 (manufacture of petroleum products). Construction is defined by codes 41 (construction of buildings), 42 (infrastructure works), and 43 (specialized construction activities). Data are from RAIS. The shaded area denotes the 2014–2016 recession period as defined by Trece et al. (2024).

Figure 3: Brazil-China Trade



*Notes:* The figure plots an index of Brazil's imports from China and exports to China (2006 = 100) between 2006 and 2018. Data are from BACI. The shaded area denotes the 2014–2016 recession period as defined by Trece et al. (2024).

ment in *Lava Jato*-sensitive sectors experienced a steeper decline during the recessions: 45% in construction and 18% in oil & gas sectors (Figure 2).

The corruption scandal was widely covered by national and international news outlets. Amid rising anti-corruption sentiment, mass protests against the federal government took place between 2015 and 2017, most notably on March 13, 2016, when between 3.6 and 6.9 million people attended protests across 337 cities (G1, 2016). President Dilma Rousseff of the *Partidos dos Trabalhadores*, PT (Workers' Party) faced declining popularity and accusations of violating fiscal responsibility law and of *Lava Jato* implications. In August 2016, President Rousseff was impeached and Vice-President Michel Temer became interim President (Agência Senado, 2016).

In this turbulent environment, municipal elections for mayors and local councilors were held in October 2016 across 5,568 cities, with compulsory voting for citizens aged 18–70. The coincidence of these elections with the trough of the recession and the unfolding *Lava Jato* investigation provides a setting to study how national economic and political shocks shape local electoral accountability.

## 2.2 Literature Review

### 2.2.1 Retrospective Voting

Retrospective voting theory models how voters reward or punish incumbent politicians based on past government performance (Healy and Malhotra, 2013). Fair (1978) was among the first to model voters' evaluation of past economic conditions within the context of U.S. presidential elections. Assuming a bipartisan system, voters choose Democrats when the expected future utility of doing so exceeds that of voting Republican. In this framework, expected future utility is a discounted function of cumulative past economic performance under Democratic administrations. A large body of research documented a positive correlation between economic performance indicators—such as GDP growth (Guntermann et al., 2021), fiscal outcomes (Brender and Drazen, 2008; Sakurai and Menezes-Filho, 2008; Drazen and Eslava, 2010) and labor market conditions (Emmenegger et al., 2015; Dassonneville et al., 2016)—and vote share of the incumbent president.

For instance, using data of U.S. presidential elections since 1948, [Guntermann et al. \(2021\)](#) found that a one percentage point increase in the election-year in GDP growth raises the incumbent’s vote share by 1.9 percentage points.

Much of this literature, however, relies on cross-sectional OLS or simple panel regressions using aggregate data, which may suffer from omitted variable bias because economic shocks are not randomly assigned ([Healy and Malhotra, 2013](#)). [Upward and Wright \(2024\)](#) addressed these issues by exploiting the variation in unexpected job losses from 1991 to 2021 using individual-level administrative data in the United Kingdom. Using a staggered differences-in-differences design, they showed that job losers are 2–3% less likely to support the incumbent party for up to 10 years.

Although a large body of literature has documented that voters punish presidents during recessions and reward them during booms, no consensus has emerged from the smaller literature on retrospective voting in local elections ([De Benedictis-Kessner and Warshaw, 2020](#)). While [Wolfers \(2002\)](#) found that the performance of U.S. states relative to the national unemployment rate is positively correlated with gubernatorial re-election, [Ebeid and Rodden \(2006\)](#) demonstrated that gubernatorial vote shares are independent of state economic conditions. Yet, the results exhibit substantial heterogeneity across economic structure: the relationship is weaker in agriculture-dominated states and stronger elsewhere ([Ebeid and Rodden, 2006](#)). Since economic structures are heterogeneous across regions and relevant to the research question, [Bartik \(1991\)](#) shift-share instruments provide a natural estimation strategy ([Borusyak et al., 2025](#)). I therefore implement a shift-share design to estimate local exposure to national shocks.

Economic shocks also shape voters’ opportunity to learn about incumbent politicians ([Ashworth et al., 2018](#)). For instance, a recession may intensify voters’ search for information about the mayor. Using a Bayesian learning framework, [Ashworth et al. \(2018\)](#) developed a model in which exogenous shocks can amplify the informativeness of signals about politician type. Empirically, [De Benedictis-Kessner and Warshaw \(2020\)](#) found suggestive evidence that newspapers affect retrospective voting in local U.S. elections, consistent with the broader literature on the role of media and political accountability (e.g., [Ferraz and Finan, 2008](#); [Snyder and Strömberg,](#)

2010). Building on this mechanism, I test whether voters were more likely to seek information about corruption during Brazil’s 2014–16 recession, and then economic voting intensified when the incumbent was associated with corruption.

### 2.2.2 Trade Shocks and Elections

A growing literature has examined the causal effects of trade shocks on political outcomes, particularly the rise of populism. In Europe, populist vote shares rose from 5% in the late 1980s to more than 20% by the 2010s (Rodrik, 2017). Populist leaders consider themselves as challengers to the status quo, claim to represent the “will of the people” against corrupt elites, and propose appealing solutions to complex economic problems (Margalit, 2019; Guriev and Papaioannou, 2022). Although the causes of populism remain debated, economic insecurity linked to globalization has emerged as a key explanation. Increased import competition from developing countries, most notably China, has harmed manufacturing and import-competing sectors in developed economies, contributing to rising domestic inequality (Guriev and Papaioannou, 2022).

A seminal contribution is Autor et al. (2013), which exploits supply-driven variation in Chinese imports to estimate their effects on U.S. labor markets. Commuting zones in the 75th percentile of exposure experienced a 4.5 percentage points decline in manufacturing employment relative to the 25th percentile. In their subsequent paper, Autor et al. (2020) examined whether the exposure to rising Chinese import competition increased political polarization. Using a shift-share instrument based on exogenous China shocks in other developed economies and variation in industry shares across commuting zones since 2000, Autor et al. (2020) found that trade-exposed districts presented higher polarization: greater support for both strong-left and strong-right political views. The mechanism seems that trade-exposed districts present higher consumption of conservative media, polarization in donation campaigns, and probability of electing conservative Republicans (Autor et al., 2020). Yet, these effects are not limited to the United States. Similarly, Colantone and Stanig (2018) showed that regions more exposed to import competition experienced an increase in support for nationalist and populist parties in 15 Western European countries over 1988–2007.

In developing countries, China operates both as a major source of imports and a significant export destination, making it possible to exploit two distinct shocks. [Costa et al. \(2016\)](#) decomposes China shocks into import supply and export demand components. They found that import supply reduced manufacturing wages by 2.9 percentage points per US\$1,000 increase in imports per worker, while export demand raised wages by 1.6 percentage points per US\$1,000 increase in exports per worker over 2000–2010. Although much of the literature focuses on developed countries, no study has explored the political implications of this decomposition in developing countries to the best of my knowledge. I plan to address this gap in the context of Brazil.

### **2.2.3 Economic Shocks and Politics in Brazil**

Much of the recent literature on trade and development economics in Brazil has exploited the process of economic liberalization between 1988 and 1995 as a source of exogenous variation for local economic conditions ([Costa et al., 2016](#)). Variation in the reduction of tariffs per industry and in the industrial share across local markets allows causal identification through shift-share designs similar to [Autor et al. \(2020\)](#). The literature suggests that regions with more tariff-exposed regions experienced delayed adjustment in wages ([Dix-Carneiro, 2014](#)), and increased crime rates and reduced public goods provision ([Dix-Carneiro et al., 2018](#)).

Trade liberalization also shaped political outcomes. Table [A1](#) summarizes recent economic papers studying the impact of economic shocks on politics in Brazil. [Ogeda et al. \(2024\)](#) demonstrated that although right-wing parties implemented the trade reforms, left-wing parties were more punished in presidential elections over the subsequent 25 years: one standard deviation decrease in tariffs is associated with a 4 percentage point decline in the vote share of left-wing presidential candidates. A plausible mechanism is that regions more affected by the liberalization experienced a weakening of labour unions, and the electoral effects are driven exclusively by parties historically linked to the union movement ([Ogeda et al., 2024](#)). Analogously, [Costa et al. \(2023\)](#) found

that regions more affected by tariffs increased the share of Pentecostal<sup>2</sup> Churches' affiliation and vote shares for Pentecostal legislative candidates by using administrative data on 161,124 religious entities between 1992 and 2014. One standard deviation decrease in tariffs increased the vote share of these candidates by 2 percentage points in legislative elections. Moreover, [Costa et al. \(2023\)](#) found that growth in Pentecostals' share contributed to the election of former right-wing populist President Jair Bolsonaro in the 2018 elections. Tariff shocks alone, however, are unlikely to fully explain the recent rise of the far right in Brazil because the severe recession of 2014–16 also increased economic uncertainty and disrupted firms ([Barros and Santos Silva, 2025](#); [Ferraz et al., 2025](#)).

Bolsonaro's election in 2018 featured a large political gender gap. [Barros and Santos Silva \(2025\)](#) estimated the effect of differential gender exposure to the 2014–2016 recession using shift-share based on industry-level employment change and regional industry composition across 551 microregions. They found that a one standard deviation increase in the difference between male and female shocks increased Bolsonaro's share by 0.6–0.9 percentage points. By using survey responses on abortion attitudes, [Barros and Santos Silva \(2025\)](#) claimed that men experiencing income losses gravitated toward Bolsonaro's right-wing rhetoric emphasizing masculine and authoritarian norms. Similarly, [Barros and Schmeißer \(2025\)](#) found that job losers are 12% more likely to join a party in two election cycles and 6%–28% more likely to run for local office. Yet, no paper has examined the impact of economic shocks on local-level politics in Brazil. The 2016 municipal election coincided with the trough of the recession, and may provide a particularly clean source of exogenous variation for a shift-share design. To the best of my knowledge, this paper is the first to study the effect of exogenous economic shocks on municipal elections in Brazil.

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<sup>2</sup>Pentecostal is a theological movement within Protestant Christianity that has been growing in Latin America in recent years.



## 3 Data

### 3.1 Elections Data

Election data are obtained from the *Tribunal Superior Eleitoral* (TSE), Brazil’s Superior Electoral Court, which provides election panel data dating back to 1945, but with higher-quality starting in the 1990s ([Tribunal Superior Eleitoral, 2024](#)). I retrieved data from *Base dos Dados*, a Brazilian NGO that provides open-source access to data. The platform removed accents in names and duplicates, replaced “9999” and “0000” with missing values, standardized municipalities to IBGE format, and harmonized other variables.

The primary dataset contains municipality- and candidate-level election results, including the election date, type of election, office contested, municipality, candidate party, electoral ID, results, and number of votes. I restricted the sample to candidates who won mayoral races in ordinary municipal elections. Brazil holds elections every two years, alternating between municipal elections every four years (e.g., 2008, 2012, 2016) and state-federal elections every four years (e.g., 2010, 2014, 2018). Beginning with the 2000 municipal elections, mayors could serve up to two consecutive four-year terms. Using the electoral ID as a key, I linked mayoral candidates in 2016 to their records in previous elections. Approximately 0.5% of observations lack an electoral ID and were excluded from the analysis.

The main dependent variable is the change in the vote share of the incumbent mayor between the 2012 and 2016 municipal elections. Conditional on the 2012 mayor not being a lame-duck, which is a candidate who had already won in 2008 and 2012, the variable is measured as the percentage of votes obtained by the incumbent mayor in the first round of the 2016 elections minus their vote share in the first round of 2012. The mean change in vote share is  $-7.20$  percentage points, implying that voters punished rather than rewarded the incumbent mayor in 2,465 municipalities, or 44% of municipalities in Brazil (Table 1).

One potential concern with this variable is that incumbent mayors who anticipate a low likelihood of winning re-election may choose not to run in 2016. Hence, those municipalities would be

dropped from the sample, and the model would miss the electoral accountability effect. Figure A1 indicates that 60% of eligible incumbent mayors ran for re-election in 2016, compared to an average of approximately 70% in other election cycles. This suggests that the recession increased the share of eligible mayors who chose not to run. If so, the estimates conditional on running may represent a lower bound on the true electoral accountability effect, since mayors who anticipated poor performance were more likely to drop out. To address this concern, I constructed an alternative variable that captures the change in the incumbent party's vote share in municipal elections.

The variable is measured as the percentage of votes obtained by the incumbent party in the first round of the 2016 elections minus their vote share in the first round of 2012. In cases where the incumbent mayor switched parties and the original party ran for re-election, the variable reflects the change in the incumbent party's vote share. The mean change in vote share of the party is  $-9.82$  percentage points, implying, on average, that voters punished rather than rewarded the incumbent party in 2,915 municipalities, or 62% of municipalities in Brazil (Table 1).

## 3.2 Employment Data

This paper constructs a shift-share variable to measure local exposure to the recession. The national shift by industry data comes from *Relação Anual de Informações Sociais* (RAIS) (Ministério do Trabalho e Emprego, 2024), and the industrial employment share by municipality comes from Census 2010 (Instituto Brasileiro de Geografia e Estatística, 2023). Both datasets are retrieved from *Base dos Dados*

The employment shift data come from RAIS, Brazil's Annual Social Information Report. It is an administrative panel employer-employee maintained by Brazil's Ministry of Labor since 1985. The dataset includes information on the employment contracts, months of hiring and separation, hours worked, wages, industry, occupation, worker demographics, and establishment size. Employers are legally required to report this information annually for all formal workers. RAIS covered 47 million formal workers in 2012 and 46 million formal workers in 2016.

Nonetheless, RAIS has three main limitations. First, it does not capture informal employ-

ment, which accounts for approximately 50% of total employment in Brazil. Although alternative datasets cover informal labor markets, RAIS is the only source available at the municipality level with annual frequency. I mitigate this issue by (i) using the employment share from Census 2010 that covers both formal and informal employment; (ii) controlling for the size of the informal labor market calculated by the Census 2010, and (iii) performing robustness checks in Appendix A regarding informality. Second, employment stock can only be calculated as of December 31. Yet, the 2016 election happened in October 2016. As a result, the 2016 employment measure is an imperfect but reasonable proxy for the pre-election year. Third, RAIS data relies on employer self-reporting and may be subject to measurement error. In addition to the recession exposure, this paper employs China trade shocks, which do not rely on the RAIS dataset.

Share employment data comes from the Census 2010 conducted by the *Instituto Brasileiro de Geografia e Estatística* (IBGE), Brazilian Institute of Geography and Statistics. The Census covered 86 million formal and informal workers in 2010 using representative sampling weighting. The micro-level individual dataset includes city, occupation, sector, employment contract type (formal or informal), and demographic characteristics (rural, male, education, welfare recipients, and income). Data for demographic controls are also from the 2010 Census. The primary sectoral identification in Brazil is the *Classificação Nacional de Atividades Econômicas* (CNAE), the National Classification of Economic Activities. Using a concordance mapping between CNAE codes in RAIS and the Census, I matched 86 2-digit industries (Figure A2).

### 3.3 Trade Data

This paper also employs a shift-share to measure local exposure to China trade shocks. The employment share comes from the Census 2010 described above. The trade shift data comes from *Base pour l'Analyse du Commerce International* (BACI), sourced from *Centre d'Études Prospectives et d'Informations Internationales* (CEPII) (CEPII, 2026). BACI reconciles bilateral and harmonized data from the United Nations Statistical Division's COMTRADE database. The dataset includes exporter/importer country and the annual value of trade in thousands of US dollars (nor-

malized to 2012 US dollars) at the 6-digit level of the Harmonized System (HS) for more than 200 countries and territories. Using concordance mapping from HS to CNAE, I matched 87 5-digit levels of CNAE industries: 18 agriculture, 7 extractive, and 62 manufacturing.

## 4 Estimaton Strategy

Two national shocks happened during 2014-16 recession: (i) recession shock driven by *Lava Jato* investigations, and (ii) the end of the commodities boom cycle. The first shock is measured by local exposure to the national recession, and the second by local exposure to the China-Brazil trade shocks.

### 4.1 Recession Exposure

Since *Lava Jato* had a heterogeneous impact on the construction and oil & gas sectors (Ferraz et al., 2025), I construct a shift-share variable measuring local recession exposure. The empirical strategy follows Bartik (1991) and, for the Brazilian context specifically, Barros and Santos Silva (2025). For municipality  $m$ , local recession exposure is defined as:

$$B_m = - \sum_{k=1}^{86} \ell_{mk,10} \cdot \Delta \log L_{k,16-12} \quad (1)$$

where the shift  $\Delta \log L_{k,16-12}$  is the log difference in national formal employment for industry  $k$  between December 31, 2016 (end of the mayoral mandate) and December 31, 2012 (beginning of the mayoral mandate), drawn from the RAIS dataset. To mitigate endogeneity, I follow the approach of leave-one-out shift from the literature by excluding the  $m$  municipality from the national employment shift (Borusyak et al., 2025). The share  $\ell_{mk,10}$  is industry  $k$ 's share of total employment (both formal and informal) in municipality  $m$ , computed from the 2010 Census for the employed population. The negative sign helps the interpretation: a higher  $B_m$  means greater recessionary exposure in municipality  $m$ . Figure A2 shows that among the five most negatively affected industries, four belong to the construction and oil & gas sectors heavily investigated by

Table 1: Summary Statistics

|                                                    | Mean   | SD      | Min     | Max       |
|----------------------------------------------------|--------|---------|---------|-----------|
| <i>Election Outcomes</i>                           |        |         |         |           |
| Δ Vote Share of Incumbent Mayor (2016-2012) (p.p.) | -7.20  | 18.35   | -81.14  | 58.32     |
| Δ Vote Share of Incumbent Party (2016-2012) (p.p.) | -9.82  | 19.13   | -81.78  | 58.32     |
| <i>Shocks</i>                                      |        |         |         |           |
| Recession Exposure                                 | 0.014  | 0.021   | -0.028  | 0.129     |
| Import Supply (per US\$1,000 imports)              | -0.060 | 0.108   | -1.590  | 0.427     |
| Export Demand (per US\$1,000 exports)              | -0.079 | 0.595   | -12.609 | 1.742     |
| <i>Electoral Control</i>                           |        |         |         |           |
| Vote share in 2012 (%)                             | 54.64  | 11.86   | 18.99   | 100.00    |
| <i>Demographic Controls</i>                        |        |         |         |           |
| Incomplete share                                   | 0.22   | 0.19    | 0.02    | 0.95      |
| Rural population share                             | 0.36   | 0.22    | 0.00    | 0.96      |
| Male population share                              | 0.50   | 0.02    | 0.46    | 0.66      |
| Non-white population share                         | 0.54   | 0.24    | 0.00    | 0.96      |
| Secondary education share                          | 0.27   | 0.07    | 0.07    | 0.50      |
| Welfare recipients share                           | 0.08   | 0.04    | 0.00    | 0.20      |
| Informal employment share                          | 0.57   | 0.15    | 0.17    | 0.89      |
| Working-age population                             | 17,691 | 125,451 | 305     | 5,549,787 |
| Average monthly income per capita (R\$)            | 475.99 | 239.97  | 106.70  | 1,999.40  |
| <i>Industry Controls</i>                           |        |         |         |           |
| Agriculture employment share                       | 0.36   | 0.19    | 0.00    | 0.87      |
| Extractive employment share                        | 0.01   | 0.02    | 0.00    | 0.27      |
| Manufacturing employment share                     | 0.10   | 0.09    | 0.00    | 0.65      |
| <i>N</i> = 2,465                                   |        |         |         |           |

*Notes:* *N* = 2,915 for incumbent party outcomes. Electoral data are from TSE. Recession exposure is the shift-share measure defined in Equation (1). Import supply and export demand are defined in Equations (3) and (4), respectively. Demographic controls and industry controls are from the 2010 Census. The incomplete share is the share of people classified as 00 (undefined) in the CNAE of the Census, following [Borusyak et al. \(2025\)](#) econometrics' recommendation.

*Lava Jato*<sup>3</sup>. These findings suggest that the recession exposure variable is indeed capturing a shock driven by *Lava Jato* investigations.

I regress the change in the incumbent mayor’s or party’s vote share between 2016 and 2012 ( $Y_{m,16-12}$ ) on local recession exposure ( $B_m$ ):

$$\Delta Y_{m,16-12} = \beta B_m + \mathbf{X}'_{m,10} \boldsymbol{\theta} + \phi_s + \varepsilon_m \quad (2)$$

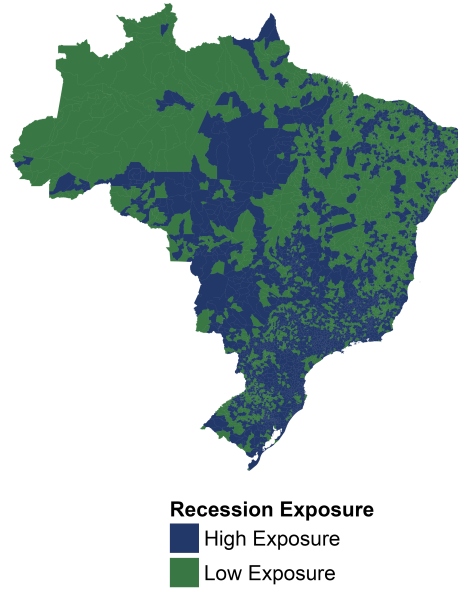
where  $\mathbf{X}'_{m,10}$  is the set of controls following [Costa et al. \(2016\)](#) and [Barros and Santos Silva \(2025\)](#). I include vote share of the incumbent in 2012 from TSE, an incomplete share control (since CNAE codes in the Census include a “00” category that not present in RAIS) following [Borusyak et al. \(2025\)](#), and demographic Census controls (rural population share, male population share, non-white population share, secondary education share, welfare recipient share, informal employment share, working-age population and a cubic polynomial of income per capita). Moreover,  $\phi_s$  denotes state fixed-effects for 27 states and robust standard errors are used.

The causal interpretation of the shift-share design relies on three main assumptions ([Borusyak et al., 2025](#)). First, there must be sufficient variation in shifts. Using 2-digit 86 industry codes allows for substantial cross-industry variation as shown in Figure [A2](#). Second, average exposure shares must be sufficiently dispersed across municipalities. I computed the Herfindahl–Hirschman Index (HHI) using the average employment shares by industry and found low concentration (HHI = 0.143). Figure [A3a](#) displays that agriculture has the highest average exposure (0.33), but Figure [A4](#) indicates that agriculture does not drive the results because the shift is close to zero. Although no specific threshold exists, average employment shares seem to be dispersed enough across municipalities. Third, shifts must be quasi-randomly assigned, implying that the industry-level shocks are uncorrelated with unobserved local determinants of political outcomes. By controlling for pre-electoral outcomes and socio-demographic characteristics that may correlate with electoral results,

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<sup>3</sup>CNAE codes: 42 (infrastructure works), 09 (mining support service activities), 41 (construction of buildings), and 71 (architecture and engineering services).

Figure 4: *Lava Jato*-driven Recession Exposure



*Notes:* The map displays the geographic distribution of local recession exposure across Brazilian municipalities. Blue municipalities are in the top 50% of recession exposure and green municipalities are in the bottom 50%. Recession exposure is the shift-share measure defined in Equation (1) using employment shifts from RAIS and employment shares from the 2010 Census.

the national shifts are plausibly orthogonal to mayoral elections. State fixed effects further absorb any time-invariant unobserved heterogeneity at the state level. Figure 4 illustrates that there is substantial within-state geographic variation in recession exposure, providing enough variation needed for identification.

Under this identification, the shift-share variable may be treated as plausibly exogenous. However, a potential concern is that unobserved local characteristics correlated with both the recession and electoral outcomes may still bias the estimates. One possible confounder is labour union strength. [Ogeda et al. \(2024\)](#) shows that Brazilian regions more affected by trade liberalization in the 1990s experienced a decline in labour union presence, which in turn reduced vote shares for left-wing parties historically linked to unions in presidential elections. Since union presence is correlated with industry-level employment shocks, it may also confound the relationship between recession exposure and municipal election outcomes. Whether labour union presence affects municipal elections remains an unanswered question. To address this concern, I implement a more

robust identification using China trade shocks as shift-share instrumental variables in the following subsection.

## 4.2 China Trade Shocks

Much of the literature of China shock on developed economies (e.g. [Colantone and Stanig, 2018](#); [Autor et al., 2020](#)) focuses exclusively on import competition exposure in local labour markets. However, the China shock has taken a distinct form in developing countries. In Brazil, the trade with China is substantial in both directions: Brazil received 14.5% of its imports by value from China and sent 15.1% of its exports to China in 2010 ([Costa et al., 2016](#)). Thus, [Costa et al. \(2016\)](#) implements two China trade shocks: (i) supply shock originating from China, the import supply shock (IS), and (ii) demand shock originating from China, the export demand shock (XD). Local China trade shocks exposures for municipality  $m$  are defined as:

$$IS_m \equiv \sum_{k=1}^{87} \ell_{mk,10} \frac{1}{L_k} \Delta I_{k,16-12} \quad (3)$$

$$XD_m \equiv \sum_{k=1}^{87} \ell_{mk,10} \frac{1}{L_k} \Delta X_{k,16-12} \quad (4)$$

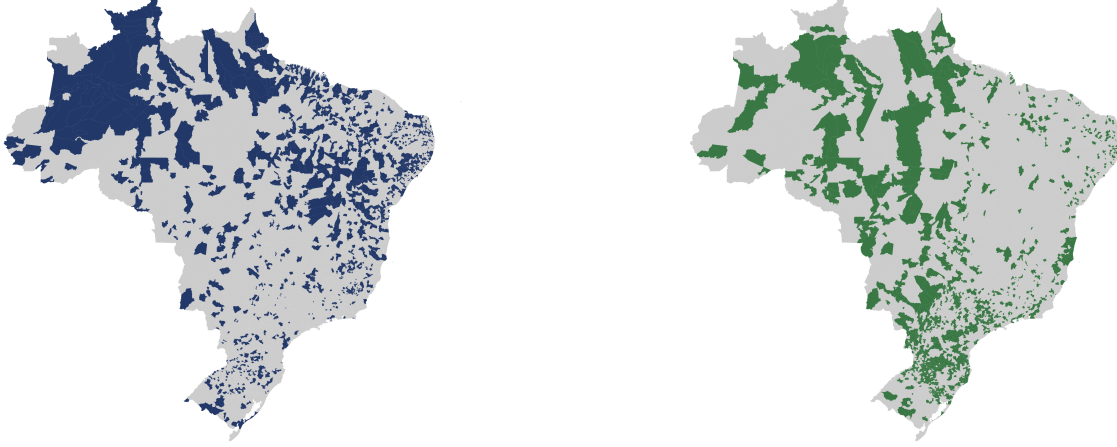
where the shifts  $\Delta I_{k,16-12}$  and  $\Delta X_{k,16-12}$  are the difference in imports/exports for industry  $k$  between 2012 and 2016 from CEPII BACI. These are divided by  $L_k$ , the total employment of industry  $k$ , to convert into imports/exports per worker in industry  $k$ , scaling the shocks relative to the industry's workforce. The share  $\ell_{mk,10}$  is industry  $k$ 's share of total employment in municipality  $m$  from the 2010 Census. Note that the industries  $k$  here differ from those in Equation (1). Following [Costa et al. \(2016\)](#), I mapped 6-digit HS product codes from BACI to 5-digit CNAE codes, matching 87 industries.

The geographic distribution of the two shocks differs substantially. Figure 5 shows that municipalities in the top 25% of import supply exposure are concentrated in the Northeast and North, while municipalities in the top 25% of export demand exposure are concentrated in the South and



Center-West. Hence, these shocks are orthogonal: correlation between IS and XD is  $-0.02$  (Figure A5a).

Figure 5: China Trade Shocks Local Exposure



(a) Top 25% Import Supply Shock

(b) Top 25% Export Demand Shock

*Notes:* Panel (a) displays municipalities in the top 25% of import supply shock exposure. Panel (b) displays municipalities in the top 25% of export demand shock exposure. Import supply and export demand shocks are defined in Equations (3) and (4), computed using trade shifts from BACI and employment shares from the 2010 Census.

However, China shocks may be correlated with Brazilian demand or supply shocks that could bias the estimates. To isolate variation originating from China, [Costa et al. \(2016\)](#) implements instrumental variables using two auxiliary regressions:

$$\frac{\Delta \tilde{I}_{ik}}{\tilde{I}_{ik,2012}} = \alpha_k + \delta_{\text{China},k} \cdot \mathbf{1}[i = \text{China}] + \mu_{ik} \quad (5)$$

$$\frac{\Delta \tilde{X}_{ik}}{\tilde{X}_{ik,2012}} = \kappa_k + \chi_{\text{China},k} \cdot \mathbf{1}[i = \text{China}] + \lambda_{ik} \quad (6)$$

where the dependent variable is the difference of values of imports/exports between 2016 and 2012 ( $\Delta \tilde{I}_{ik}$ ,  $\Delta \tilde{X}_{ik}$ ) divided by base values of imports/exports values in 2012 ( $\tilde{I}_{ik,2012}$ ,  $\tilde{X}_{ik,2012}$ ) for country  $i$  and industry  $k$ , excluding Brazil to mitigate endogeneity. The sector fixed effects ( $\alpha_k$ ,  $\kappa_k$ ) capture the mean growth rate of each sector net of Brazil, and absorb any world-level

shocks. The regressions are weighted by the size of imports/exports, so that the results are not driven by large growth rates in small countries. The coefficient  $\delta_{\text{China},k}$  (or  $\chi_{\text{China},k}$ ) captures how much China's trade demand (or supply) with all other countries except Brazil evolved differently from the weighted average country in sector  $k$ .

I then construct the shift variable of the instrumental variable for export demand as the Brazilian exports to China in 2012 multiplied by China's trade demand elasticity from the auxiliary regression:  $\widehat{\Delta X}_{k,16-12} \equiv X_{k,2012} \cdot \hat{\delta}_{\text{China},k}$ . The instrumental variable for import supply is the Brazilian imports from China in 2012 multiplied by China's trade supply elasticity from the auxiliary regression:  $\widehat{\Delta I}_{k,16-12} \equiv I_{k,2012} \cdot \hat{\chi}_{\text{China},k}$ . Both instrumental variables for municipality  $m$  are defined as:

$$ivIS_m \equiv \sum_{k=1}^{87} \ell_{mk,10} \frac{1}{L_k} \widehat{\Delta I}_{k,16-12} \quad (7)$$

$$ivXD_m \equiv \sum_{k=1}^{87} \ell_{mk,10} \frac{1}{L_k} \widehat{\Delta X}_{k,16-12} \quad (8)$$

The main regression estimates the impact of  $IS$  and  $XD$  on the change in the incumbent's vote share.

$$\Delta Y_{m,16-12} = \zeta_I IS_m + \zeta_X XD_m + \beta B_m + \mathbf{T}'_{m,10} \boldsymbol{\theta} + \phi_s + \varepsilon_m \quad (9)$$

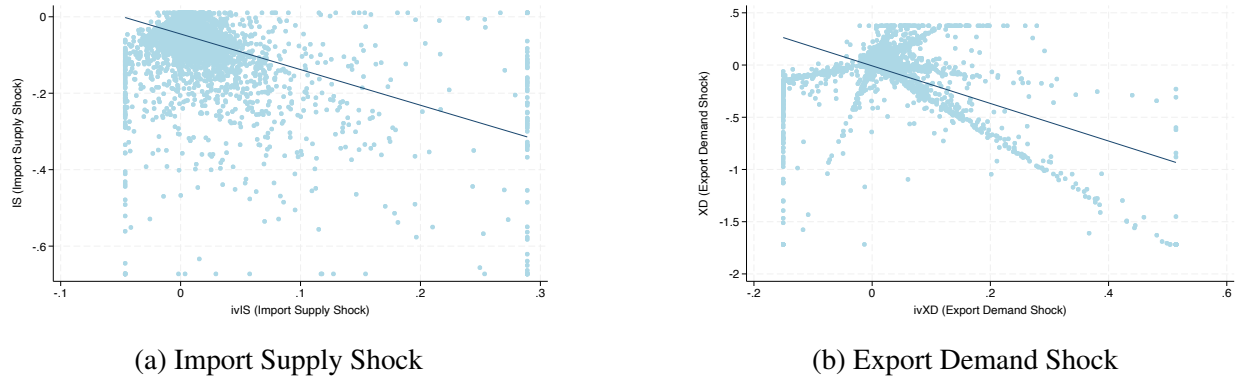
where  $\mathbf{T}'_{m,10}$  is the set of controls described above,  $\mathbf{X}'_{m,10}$  plus agriculture, extractive, and manufacturing shares from the 2010 Census, so the estimates are identified within cities with the same initial exposure to Chinese trade. Some specifications control for the local recession exposure  $B_m$ . Figures A5c and A5b illustrate that the recession shock is orthogonal to  $XD$  (correlation of  $-0.01$ ) and moderately correlated with  $IS$  (correlation of  $-0.48$ ).

All regressions include state fixed effects ( $\phi_s$ ) to absorb any time-invariant state-level shocks, and are weighted by each municipality's share of the national workforce, so that the estimates are interpretable as effects per worker. The trade shock variables  $IS_m$ ,  $XD_m$ , and their instruments

$ivIS_m$  and  $ivXD_m$  are winsorized at the 1st and 99th percentiles to mitigate the influence of outliers. I also use robust standard errors.

The causal identification of an instrumental variable relies on two main assumptions. The first is relevance: the instrument must be strongly correlated with endogenous variables. Figures 6 reports that  $ivIS$  is correlated with  $IS$  ( $-0.53$ ) and  $ivXD$  is correlated with  $XD$  ( $-0.54$ ). Furthermore, Kleibergen-Paap rk Wald F-statistic is above 10 in the main regression tables in the following section. The second condition is exogeneity:  $ivXD$  and  $ivIS$  must be uncorrelated with municipal election outcomes. In other words, how much China trades with all countries except Brazil has evolved differently from the weighted average country in each sector must be uncorrelated with local determinants of municipal elections. This assumption seems feasible because municipal elections are unlikely to influence China's bilateral trade patterns with other countries.

Figure 6: Correlation between Raw Variables and Instrumental Variables



Notes: Panel (a) plots the import supply shock ( $IS$ ) against its instrumental variable ( $ivIS$ ). Panel (b) plots the export demand shock ( $XD$ ) against its instrumental variable ( $ivXD$ ). Each observation is a municipality. The fitted line is from a linear regression. Correlations are  $-0.53$  for  $IS$  and  $ivIS$ , and  $-0.54$  for  $XD$  and  $ivXD$ . Variables are winsorized at the 1st and 99th percentiles.

## 5 Results

First, I present the results of two constructed shocks, *Lava Jato* and Chinese trade, on the 2016 municipal elections. Second, I examine whether voters held the presidential party accountable for the same shocks in the 2018 presidential election.

## 5.1 Municipal Elections

First, I estimate the effect of recession exposure on municipal electoral outcomes, where the recession exposure variable is most sensitive to *Lava Jato*-investigated sectors. Using Equation (2), Table 2 reports the coefficient of local exposure on the change of the incumbent's vote share. In Panel A, the dependent variable is the change in the vote share of the incumbent mayor. Across columns (1)–(4), the coefficients are statistically insignificant. The point estimate in column (1) is positive, but turns negative once I control for the 2012 vote share in column (2), suggesting that the initial positive estimate reflects mean reversion. Although the negative coefficients in columns (2)–(4) are not significant, their sign is consistent with voters punishing incumbents in municipalities more exposed to the recession. In Panel B, the dependent variable is the change in the vote share of the incumbent's party. In column (2), the coefficient is  $-39.19$ , and is statistically significant. Municipalities at the 75th percentile of recession exposure decrease the incumbent vote share by 1.06 percentage points ( $-39.19 \times 0.027$ ) compared to those at the 25th percentile. Yet, once demographic controls, mainly informality share, coefficients in columns (3) and (4) lose their significance. Accordingly, voters do not systematically punish the incumbent for exposure to the recession driven by *Lava Jato*.

Furthermore, I perform robustness checks in Table A2. Column (1) switches the dependent variable for the coalition vote share, defined as the change in the vote share of the incumbent party, whether it runs independently or as part of a coalition, and finds no significant effect of recession exposure. Column (2) redefines the shift-share period to begin on December 31, 2013, right before the start of the recession in the first quarter of 2014, and again finds no significant effects. Additionally, Column (3) implements the level difference rather than the log difference, and the coefficient of recession exposure remains insignificant.

Since the employment shift data covers only formal employment (RAIS), I divide the sample into municipalities below and above the median informal employment share from the 2010 Census. If voters punish mayors for the recession but RAIS fails to capture informal sector job losses, I would expect a more pronounced effect in cities below the informality median, where formal

Table 2: Effect of Recession Exposure on Incumbent Mayor's Vote Share

|                                                                | (1)                | (2)                 | (3)              | (4)               |
|----------------------------------------------------------------|--------------------|---------------------|------------------|-------------------|
| <i>Panel A: <math>\Delta</math> Incumbent Mayor Vote Share</i> |                    |                     |                  |                   |
| Recession Exposure ( <i>Lava Jato</i> )                        | 21.53<br>(18.19)   | -11.00<br>(16.89)   | -2.01<br>(21.62) | -10.25<br>(22.25) |
| <i>N</i>                                                       | 2,466              | 2,466               | 2,465            | 2,465             |
| <i>Panel B: <math>\Delta</math> Incumbent Party Vote Share</i> |                    |                     |                  |                   |
| Recession Exposure ( <i>Lava Jato</i> )                        | -29.45*<br>(16.94) | -39.19**<br>(15.83) | 7.07<br>(20.06)  | 2.79<br>(20.45)   |
| <i>N</i>                                                       | 2,915              | 2,915               | 2,914            | 2,914             |
| Electoral Control                                              |                    | ✓                   | ✓                | ✓                 |
| Demographic Controls                                           |                    |                     | ✓                | ✓                 |
| State Fixed Effects                                            |                    |                     |                  | ✓                 |
| IQR of Recession Exposure                                      | 0.027              | 0.027               | 0.027            | 0.027             |

*Notes:* Dependent variable is the change in vote share (p.p.) between the 2016 and 2012 elections. Panel A uses the incumbent mayor's vote share and Panel B uses the incumbent party's vote share. IQR (interquartile range) is the difference between the 75th and 25th percentiles of recession exposure. Electoral controls include the incumbent's vote share in 2012. Demographic controls include controls from Table 1. State fixed effects are for 27 Brazilian states. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

employment better represents the local labour market. Yet, Table 3 shows no such clear pattern. These results suggest that potential measurement error in RAIS does not meaningfully affect the estimates.

Table 3: Effect of Recession Exposure by Informality

|                                                                | (1)<br>Below Median | (2)<br>Above Median |
|----------------------------------------------------------------|---------------------|---------------------|
| <i>Panel A: <math>\Delta</math> Incumbent Mayor Vote Share</i> |                     |                     |
| Recession Exposure                                             | 12.94<br>(30.00)    | -34.17<br>(34.55)   |
| <i>N</i>                                                       | 1,218               | 1,247               |
| <i>Panel B: <math>\Delta</math> Incumbent Party Vote Share</i> |                     |                     |
| Recession Exposure                                             | 16.67<br>(27.47)    | 44.53<br>(35.59)    |
| <i>N</i>                                                       | 1,471               | 1,443               |
| Electoral Control                                              | ✓                   | ✓                   |
| Demographic Controls                                           | ✓                   | ✓                   |
| State Fixed Effects                                            | ✓                   | ✓                   |
| IQR of Recession Exposure                                      | 0.033               | 0.014               |

*Notes:* Dependent variable is the change in vote share (p.p.) between the 2016 and 2012 elections. The sample is split at the median of the informal employment share from the 2010 Census. Panel A uses the incumbent mayor's vote share and Panel B uses the incumbent party's vote share. IQR is the interquartile range, the difference between the 75th and 25th percentiles of recession exposure within each subsample. All specifications include electoral controls, demographic controls, and state fixed effects. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

Second, I estimate the effect of China trade shocks on municipal elections. [Costa et al. \(2016\)](#) shows that import-competing regions experienced slower wage growth while commodity-exporting regions experienced faster wage growth between 2000 and 2010. As a preliminary check, I test whether China trade shocks directly impacted local labour markets by regressing the change in formal employment share<sup>4</sup> on import-supply (*IS*) and export demand (*XD*). Table 4 presents the coefficients from Equation (9), excluding recession exposure and electoral controls, and including

<sup>4</sup>Number of formal workers divided by total number of workers

the change in formal employment share between 2008 and 2012 as a pre-trends control.

Across columns (4)–(6) that implement the IV specification, all coefficients on  $IS$  and  $XD$  are statistically significant. The preferred specification is column (6), which includes state fixed-effects. Export demand increased the formal employment share by 1.4 percentage points per US\$1,000 increase in exports per worker. Meanwhile, import supply increased the formal employment share by 7.6 percentage points per US\$1,000 increase in imports per worker. Also, [Costa et al. \(2016\)](#) finds similar coefficients of trade shocks on total employment share.

The interpretation of export demand shocks on formal employment share is straightforward: cities that experienced an exogenous China demand shock saw greater economic activity and, consequently, higher employment. Yet, the impact of import supply on employment is more nuanced. [Costa et al. \(2016\)](#) argues that employment may increase within the household in a backward-bending labour supply curve. For example, if one manufacturing worker loses their job due to Chinese import competition, other household members might enter the formal labour market to maintain household consumption. Regardless of whether the explanation is fully convincing, the results confirm that China trade shocks affected Brazil's local labor markets between 2012 and 2016. Yet, the substantial difference in the magnitude of the two coefficients remains an open question beyond the scope of this thesis.

Having established that China trade shocks significantly affected local labour markets, I now test whether these shocks translated into electoral accountability at the municipal level. Table 5 presents the coefficient of Equation (9) on the change of incumbent mayor's vote share. Across columns (1)–(5), none of the coefficients of import supply, export demand, and recession exposure are statistically significant. In column (5), the preferred specification, the point coefficient of  $IS$  is negative, which is consistent with voters punishing mayors for Chinese import penetration, and  $XD$  is positive, which is consistent with voters rewarding mayors for Chinese demand, though neither is significant.

I further examine whether the China trade shocks influenced local electoral accountability by using the incumbent party's vote share as an alternative outcome variable (Table A3), and repli-

Table 4: Effect of China Trade Shocks on Formal Employment

|                                                               | (1)                 | (2)              | (3)              | (4)                 | (5)                | (6)                 |
|---------------------------------------------------------------|---------------------|------------------|------------------|---------------------|--------------------|---------------------|
| <i>Dep. Var.: <math>\Delta</math> Formal Employment Share</i> |                     |                  |                  |                     |                    |                     |
| Import Supply (IS)                                            | 0.089***<br>(0.017) | 0.019<br>(0.014) | 0.016<br>(0.015) | 0.124***<br>(0.029) | 0.095**<br>(0.041) | 0.105***<br>(0.038) |
| Export Demand (XD)                                            | 0.025**<br>(0.010)  | 0.006<br>(0.006) | 0.005<br>(0.006) | 0.042***<br>(0.012) | 0.017**<br>(0.008) | 0.014*<br>(0.007)   |
| KP F-stat                                                     |                     |                  |                  | 64.60               | 37.93              | 38.35               |
| <i>N</i>                                                      | 5,511               | 5,508            | 5,508            | 5,511               | 5,508              | 5,508               |
| Estimation                                                    | OLS                 | OLS              | OLS              | IV                  | IV                 | IV                  |
| Pre-Trends Control                                            |                     | ✓                | ✓                |                     | ✓                  | ✓                   |
| Industry Control                                              |                     | ✓                | ✓                |                     | ✓                  | ✓                   |
| Demographic Controls                                          |                     | ✓                | ✓                |                     | ✓                  | ✓                   |
| State Fixed Effects                                           |                     |                  | ✓                |                     |                    | ✓                   |

*Notes:* Dependent variable is the change in formal employment share from 2012 to 2016, constructed from RAIS. Columns (1)–(3) report OLS estimates. Columns (4)–(6) report IV estimates instrumenting *IS* and *XD* with shift-share instruments based on China’s bilateral trade with all countries excluding Brazil. Pre-trends control is the change in formal employment share between 2008 and 2012. Industry controls include agriculture, extractive, and manufacturing employment shares from the 2010 Census. All regressions are weighted by each municipality’s share of the national workforce for interpretation of the coefficient as a US\$1000 increase in exports/imports per worker. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .



Table 5: Effect of China Trade Shocks on Incumbent Mayor Vote Share

|                                                                  | (1)             | (2)              | (3)              | (4)              | (5)               |
|------------------------------------------------------------------|-----------------|------------------|------------------|------------------|-------------------|
| <i>Dep. Var.: <math>\Delta</math> Incumbent Mayor Vote Share</i> |                 |                  |                  |                  |                   |
| Import Supply (IS)                                               | 10.73<br>(8.00) | 10.39<br>(8.12)  | -1.34<br>(24.74) | -4.69<br>(25.32) | -5.61<br>(26.33)  |
| Export Demand (XD)                                               | 1.25<br>(2.57)  | 1.31<br>(2.53)   | 3.80<br>(3.38)   | 0.49<br>(3.28)   | 0.37<br>(3.26)    |
| Recession Exposure ( <i>Lava Jato</i> )                          |                 | 12.84<br>(56.13) |                  |                  | -19.95<br>(70.85) |
| KP F-stat                                                        |                 |                  | 19.45            | 18.79            | 17.53             |
| <i>N</i>                                                         | 2,465           | 2,465            | 2,465            | 2,465            | 2,465             |
| Estimation                                                       | OLS             | OLS              | IV               | IV               | IV                |
| Electoral Controls                                               | ✓               | ✓                | ✓                | ✓                | ✓                 |
| Industry Controls                                                | ✓               | ✓                | ✓                | ✓                | ✓                 |
| Demographic Controls                                             | ✓               | ✓                | ✓                | ✓                | ✓                 |
| State Fixed Effects                                              | ✓               | ✓                |                  | ✓                | ✓                 |

*Notes:* Dependent variable is the change in the incumbent mayor's vote share (p.p.) between the 2016 and 2012 elections. Columns (1)–(2) report OLS estimates. Columns (3)–(5) report IV estimates instrumenting *IS* and *XD* with shift-share instruments based on China's bilateral trade with all countries excluding Brazil. Electoral controls include the incumbent's vote share in 2012. Industry controls include agriculture, extractive, and manufacturing employment shares from the 2010 Census. Demographic controls are described in Table 1. All regressions are weighted by each municipality's share of the national workforce. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

cating [Autor et al. \(2020\)](#)'s methodology (Table A4). In both analyses, the coefficients of China trade shocks remain insignificant. Given the importance of informality in Brazilian labour markets, Table A5 splits the sample into below and above the informality median and finds no heterogeneity, though the F-statistics are lower. Since China shocks impacted agricultural and manufacturing regions differently ([Costa et al., 2016](#)), Table A6 presents a sub-sample analysis for more and less manufacturing-intensive and agricultural-intensive regions. Despite very low F-statistics in some cases, there are no significant results.

While China trade shocks significantly impacted local labor markets between 2012 and 2016, they did not translate into electoral accountability effects in the 2016 municipal elections. Since neither local recession exposure driven by *Lava Jato* nor China trade shocks impacted municipal elections on average, there is suggestive evidence that voters exercise a degree of sophistication in attributing responsibility. The average voter seems to recognize that mayoral responsibilities are limited to areas such as primary education, primary healthcare, and local infrastructure, and hence does not hold mayors accountable for national recessions or shifts in China-Brazil bilateral trade.

Another possibility is that voters might hold specifically the left-wing presidential party, *Partido dos Trabalhadores* (PT), responsible for the national corruption scandal and China trade disruptions. As described in Section 2, mass protests took place in 2016 against the PT federal government amid the *Lava Jato* scandal, and voters may have extended this sentiment to PT mayors in the concurrent municipal elections. Column (1) of Table 6 reports that neither China trade shocks nor recession exposure drove electoral punishment for PT incumbents. To address the concern that PT mayors elected in 2012 may have switched parties in anticipation of a re-election loss, I estimate the effect of the shocks on the likelihood of party switching conditional on having run for PT in 2012, and find no significant effects (Table A7). Therefore, the average voter did not punish PT incumbents in the 2016 municipal elections, reinforcing the hypothesis that voters exercise sophistication in attributing accountability across levels of government.

Amid rising anti-left sentiment, [Barros and Santos Silva \(2025\)](#) shows the electoral rise of right-wing former President Jair Bolsonaro in 2018. I test whether voters punished left-wing parties

more broadly. Columns (2) and (3) of Table 6 present results for mayors from left-wing parties excluding PT, using a comprehensive and a restricted ideology classification, respectively.<sup>5</sup> Both specifications yield similar and significant coefficients on recession exposure. In the preferred specification (column 3), municipalities at the 75th percentile of recession exposure decreased the incumbent vote share by 10.39 percentage points ( $-399.82 \times 0.026$ ) relative to those at the 25th percentile, when the incumbent belonged to a non-PT left-wing party. While the average voter did not punish PT incumbents, they did punish other left-wing incumbents. One possible explanation is that PT, as the dominant left-wing party, controlled greater resources and incumbency advantages than other left-wing parties.

I further test whether the shocks shaped electoral accountability for right-wing and center parties. Table A8 reports that municipalities at the 75th percentile of import supply exposure increased the incumbent vote share by 3.10 percentage points more than those at the 25th percentile when the incumbent was from a right-wing party. This suggests that voters in municipalities more exposed to Chinese import competition rewarded right-wing incumbents, consistent with Autor et al. (2020), which establishes increased support for conservative candidates in trade-exposed districts in the United States.

In summary, voters did not punish incumbent mayors or parties for the *Lava Jato*-driven recession or China trade shocks on average. While voters did not hold PT incumbents accountable, they did punish other left-wing incumbents for recession exposure, and import supply shocks benefited right-wing incumbents. This heterogeneity is consistent with Barros and Santos Silva (2025), who find the rise of right-wing politicians such as Bolsonaro in the 2018 presidential election as a response to the 2014–2016 recession.

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<sup>5</sup>The restricted classification defines left-wing parties as those with an ideology score below  $-0.2$  on a scale from  $-1$  to  $1$  based on Zucco and Power (2023). The comprehensive classification additionally includes minor parties outside Zucco and Power (2023), classified using Wikipedia.

Table 6: Effect of Shocks on Incumbent Mayor Vote Share Across Left-Wing

|                                                                  | (1)<br>PT Incumbent | (2)<br>Left (non-PT)  | (3)<br>Left (non-PT)  |
|------------------------------------------------------------------|---------------------|-----------------------|-----------------------|
| <i>Dep. Var.: <math>\Delta</math> Incumbent Mayor Vote Share</i> |                     |                       |                       |
| Import Supply (IS)                                               | −4.01<br>(112.94)   | −97.03<br>(92.81)     | −96.66<br>(92.53)     |
| Export Demand (XD)                                               | 0.72<br>(12.74)     | 2.17<br>(4.56)        | 1.61<br>(4.69)        |
| Recession Exposure ( <i>Lava Jato</i> )                          | 79.35<br>(382.67)   | −395.98**<br>(198.29) | −399.82**<br>(198.84) |
| KP F-stat                                                        | 0.94                | 3.07                  | 3.07                  |
| <i>N</i>                                                         | 295                 | 495                   | 491                   |
| Ideology Classification                                          |                     | Comprehensive         | Restricted            |
| Electoral Controls                                               | ✓                   | ✓                     | ✓                     |
| Industry Controls                                                | ✓                   | ✓                     | ✓                     |
| Demographic Controls                                             | ✓                   | ✓                     | ✓                     |
| State Fixed Effects                                              | ✓                   | ✓                     | ✓                     |
| IQR of Recession Exposure                                        | 0.028               | 0.026                 | 0.026                 |
| IQR of Import Supply                                             | 0.057               | 0.043                 | 0.044                 |
| IQR of Export Demand                                             | 0.030               | 0.020                 | 0.021                 |

*Notes:* Dependent variable is the change in the incumbent mayor's vote share (p.p.) between the 2016 and 2012 elections. Each column restricts the sample to municipalities where the incumbent mayor belongs to the indicated party group. Column (1) restricts to PT incumbents. Columns (2) and (3) restrict to left-wing non-PT incumbents using a comprehensive and restricted ideology classification, respectively, which is explained in footnote 5. All columns report IV estimates instrumenting *IS* and *XD* with shift-share instruments based on China's bilateral trade with all countries excluding Brazil. Electoral controls include the incumbent's vote share in 2012. Industry controls include agriculture, extractive, and manufacturing employment shares from the 2010 Census. Demographic controls are described in Table 1. IQR is the interquartile range, which refers to the difference between the 75th and 25th percentiles of each shock. All regressions are weighted by each municipality's share of the national workforce. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

## 5.2 Presidential Elections in 2018

If voters do not hold mayors accountable for national shocks, they might hold the president accountable in the subsequent presidential election in 2018. Hence, I estimate the effect of the recession exposure on the change in PT’s presidential vote share between 2014 and 2018.<sup>6</sup>

Table 7 reports the coefficient on *Lava Jato*-driven recession exposure over the 2014–2018 period (presidential mandate) using Equation (2). Panel A calculates the vote share from the first round of presidential elections, and Panel B from the second round of elections. Across all columns and both panels, the coefficients are statistically significant. The preferred specification is column (4), which includes pre-election and demographic controls as well as state fixed effects. In the first round, municipalities at the 75th percentile of recession exposure punished PT by 2.60 percentage points more than those at the 25th percentile ( $-12.59 \times 0.207$ ). In the second round, the corresponding estimate is 3.21 percentage points ( $-15.52 \times 0.207$ ). As a robustness check, I change the recession window to 2012–2016, matching the mayoral election cycle, and find statistically significant results with lower magnitudes (Table A9). Overall, there is convincing evidence that voters held the president accountable for the *Lava Jato* shock.

Furthermore, I test whether China trade shocks influenced presidential elections. Table ?? presents the coefficients from Equation (9). Across columns (3)–(5), which implement the IV specification, the coefficients on import supply are statistically significant and negative. The preferred specification is column (5), which additionally controls for recession exposure. The coefficient on recession exposure remains significant and similar in magnitude to that in Table 7. In the first round (Panel A), municipalities at the 75th percentile of import supply punished PT by 0.47 percentage points more than those at the 25th percentile ( $-11.65 \times 0.041$ ). In the second round (Panel B), the corresponding estimate is 0.64 percentage points ( $-15.66 \times 0.041$ ). Since regions more affected by Chinese import competition experienced slower manufacturing wage growth (Costa et

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<sup>6</sup>PT (*Partido dos Trabalhadores*, Workers’ Party) has been the major left-wing party in Brazil, holding the presidency from 2003 to 2016 until former President Dilma Rousseff was impeached. In 2018, the party ran Fernando Haddad as its presidential candidate. The change in vote share is therefore calculated as the difference between Haddad’s vote share in 2018 and Rousseff’s vote share in 2014.

Table 7: Effect of Recession Exposure on PT Presidential Vote Share

|                                                                        | (1)                 | (2)                 | (3)                 | (4)                 |
|------------------------------------------------------------------------|---------------------|---------------------|---------------------|---------------------|
| <i>Panel A: <math>\Delta</math> PT Vote Share, Round 1 (2018–2014)</i> |                     |                     |                     |                     |
| Recession Exposure <sub>14–18</sub>                                    | −9.95***<br>(0.74)  | 13.22***<br>(0.76)  | −26.00***<br>(2.19) | −12.59***<br>(1.63) |
| <i>N</i>                                                               | 5,565               | 5,565               | 5,565               | 5,565               |
| <i>Panel B: <math>\Delta</math> PT Vote Share, Round 2 (2018–2014)</i> |                     |                     |                     |                     |
| Recession Exposure <sub>14–18</sub>                                    | −17.07***<br>(0.49) | −16.69***<br>(0.53) | −18.58***<br>(1.81) | −15.52***<br>(1.74) |
| <i>N</i>                                                               | 5,565               | 5,565               | 5,565               | 5,565               |
| Electoral Control                                                      |                     | ✓                   | ✓                   | ✓                   |
| Demographic Controls                                                   |                     |                     | ✓                   | ✓                   |
| State Fixed Effects                                                    |                     |                     |                     | ✓                   |
| IQR of Recession Exposure                                              | 0.207               | 0.207               | 0.207               | 0.207               |

*Notes:* Dependent variable is the change in PT's presidential vote share (p.p.) between the 2018 and 2014 elections. Panel A uses the first round and Panel B uses the second round. IQR is the interquartile range, refers to the difference between the 75th and 25th percentiles of a distribution. Electoral controls include the 2010–2014 change in PT's presidential vote share. Demographic controls are described in Table 1. State fixed effects are indicator variables for each of the 27 Brazilian states. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

al., 2016), voters in those regions punished the president on average.

For export demand, the coefficient in column (5) is negative and significant in the first round (Panel A), but not in the second round (Panel B). In the first round, municipalities at the 75th percentile of export demand rewarded PT by 0.17 percentage points more than those at the 25th percentile ( $2.72 \times 0.061$ ). Since regions more exposed to Chinese demand experienced wage and economic activity growth (Costa et al., 2016), voters rewarded the president accordingly. However, the magnitude of punishment due to import supply exceeds the magnitude of reward from export demand, suggesting that voters may hold the president asymmetrically accountable for China trade shocks.

As a robustness check, I change the China shock window to 2012-2016, matching the mayoral election cycle, and find similar results with lower significance and magnitude (Table A10). Taken together, the municipal and presidential results provide consistent evidence that voters exercise sophistication in attributing responsibility across levels of government: they did not hold mayors accountable for national shocks, but did punish the president.

This interpretation is further supported by the transience of party affiliation at the local level in Brazil. Because mayors and candidates constantly switch parties, party labels are a weak signal of federal government affiliation. Figure A6 displays that in 2016, 47% of incumbent mayors and 54% of mayors' candidates ran under a different party than in 2012, making it difficult for voters to use party affiliation as a proxy for federal accountability.

## 6 Information Mechanism

Access to information substantially shapes electoral accountability (Snyder and Strömberg, 2010). For instance, Ferraz and Finan (2008) found that electoral accountability for corruption is more pronounced in Brazilian municipalities with greater local media access. This section tests whether economic voting driven by the *Lava Jato* recession is amplified by access to information.

First, I study the role of general information. Using the 2010 Census, I calculate the share of households with access to cell phones and computers with internet at the city level. Figure 7

Table 8: Effect of China Trade Shocks on PT Presidential Vote Share

|                                                                        | (1)               | (2)                 | (3)                 | (4)                 | (5)                 |
|------------------------------------------------------------------------|-------------------|---------------------|---------------------|---------------------|---------------------|
| <i>Panel A: <math>\Delta</math> PT Vote Share, Round 1 (2018–2014)</i> |                   |                     |                     |                     |                     |
| Import Supply (IS)                                                     | 2.31**<br>(1.16)  | 1.99<br>(1.16)      | −26.32***<br>(6.86) | −12.64**<br>(4.95)  | −11.65**<br>(4.86)  |
| Export Demand (XD)                                                     | 1.51***<br>(0.42) | 1.39***<br>(0.42)   | 0.73<br>(1.29)      | 3.14***<br>(1.00)   | 2.72***<br>(1.01)   |
| Recession Exposure                                                     |                   | −7.98***<br>(1.98)  |                     |                     | −8.95***<br>(2.10)  |
| KP F-stat                                                              |                   |                     | 27.13               | 26.05               | 26.51               |
| <i>N</i>                                                               | 5,564             | 5,564               | 5,564               | 5,563               | 5,563               |
| <i>Panel B: <math>\Delta</math> PT Vote Share, Round 2 (2018–2014)</i> |                   |                     |                     |                     |                     |
| Import Supply (IS)                                                     | 0.52<br>(1.35)    | −0.07<br>(1.34)     | −24.36***<br>(6.79) | −16.93***<br>(5.93) | −15.66***<br>(5.84) |
| Export Demand (XD)                                                     | 1.10**<br>(0.43)  | 0.88**<br>(0.43)    | −3.83***<br>(1.40)  | 0.06<br>(1.16)      | −0.65<br>(1.17)     |
| Recession Exposure                                                     |                   | −12.03***<br>(2.05) |                     |                     | −14.33***<br>(2.23) |
| KP F-stat                                                              |                   |                     | 27.08               | 26.02               | 26.48               |
| <i>N</i>                                                               | 5,564             | 5,564               | 5,564               | 5,563               | 5,563               |
| Estimation                                                             | OLS               | OLS                 | IV                  | IV                  | IV                  |
| Electoral Controls                                                     | ✓                 | ✓                   | ✓                   | ✓                   | ✓                   |
| Industry Controls                                                      | ✓                 | ✓                   | ✓                   | ✓                   | ✓                   |
| Demographic Controls                                                   | ✓                 | ✓                   | ✓                   | ✓                   | ✓                   |
| State Fixed Effects                                                    | ✓                 | ✓                   |                     | ✓                   | ✓                   |
| IQR of Import Supply                                                   | 0.041             | 0.041               | 0.041               | 0.041               | 0.041               |
| IQR of Export Demand                                                   | 0.061             | 0.061               | 0.061               | 0.061               | 0.061               |
| IQR of Recession Exposure                                              |                   | 0.207               |                     |                     | 0.207               |

*Notes:* Dependent variable is the change in PT's presidential vote share (p.p.) between the 2018 and 2014 elections. Panel A uses the first round and Panel B uses the second round. Columns (1)–(2) report OLS estimates. Columns (3)–(5) report IV estimates instrumenting *IS* and *XD* with shift-share instruments based on China's bilateral trade with all countries excluding Brazil. Electoral controls include the 2010–2014 change in PT's presidential vote share. Industry and demographic controls are described in Table 1. IQR is the interquartile range, which refers to the difference between the 75th and 25th percentiles of each shock. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

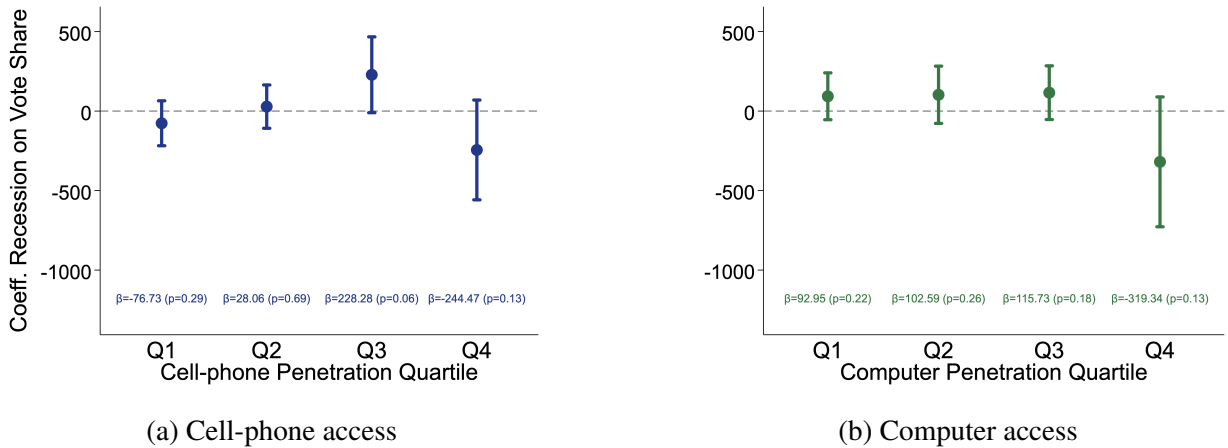


plots the coefficient of recession exposure on the change in the incumbent mayor's vote share from Equation (9) across quartiles of cell-phone and computer penetration.

Figure 7a displays that municipalities in the bottom 50% of cell-phone penetration do not punish mayors for the recession, and those in the third quartile present the opposite sign. Yet, among municipalities in the top 25% of cell-phone penetration, those at the 75th percentile of recession exposure punish incumbent mayors by 8.31 percentage points more ( $-244.47 \times 0.034$ ) than those at the 25th percentile. However, this coefficient is not statistically significant ( $p = 0.13$ ), likely due to the small number of observations.

Similarly, Figure 7b shows that municipalities in the bottom 75% of computer penetration do not punish mayors for the recession. Yet, among municipalities in the top 25%, those at the 75th percentile of recession exposure punish incumbent mayors by 9.58 percentage points more ( $-319.34 \times 0.030$ ) than those at the 25th percentile. Again, this coefficient is not statistically significant ( $p = 0.13$ ).

Figure 7: Effect of Recession Exposure on Vote Share by Media Penetration Quartile



*Notes:* Each point displays the coefficient of recession exposure on the change in the incumbent mayor's vote share from Equation (9), estimated separately for each quartile of media penetration. Panel (a) uses cell-phone access and Panel (b) uses computer with internet access, both measured as the share of households with access from the 2010 Census. Number of observations per quartile is 616. Vertical bars represent 95% confidence intervals.  $p$ -values are reported below each coefficient. Sample sizes are reported for each quartile.

Although the results are not statistically significant and non-monotonic, the magnitudes at the top quartile are suggestive of an information channel. Since cell-phone and computer penetra-

tion are broad proxies for information access, I next turn to measures more specific to the *Lava Jato* corruption scandal and the mass protests that occurred concurrently with the 2016 municipal elections.

*Lava Jato* affected almost all major parties.<sup>7</sup> I estimate the effect of recession exposure on the incumbent mayor's vote share across parties with different numbers of *Lava Jato* defendants, based on [Oliveira and Teixeira \(2022\)](#). The hypothesis is that if economic shocks increase voters' incentive to seek information about the incumbent ([Ashworth et al., 2018](#); [De Benedictis-Kessner and Warshaw, 2020](#)), mayors belonging to parties under investigation should face greater electoral punishment for the recession than those from other parties.

Figure 8 presents the recession exposure coefficient from Equation (9) across four subsamples: (i) incumbent mayors from parties with no *Lava Jato* defendants ("No Defendants"); (ii) those from parties with at least one defendant ("Any Defendants"); (iii) those from parties with more than five defendants ("5+ Defendants"); and (iv) those from parties with more than ten defendants ("10+ Defendants").

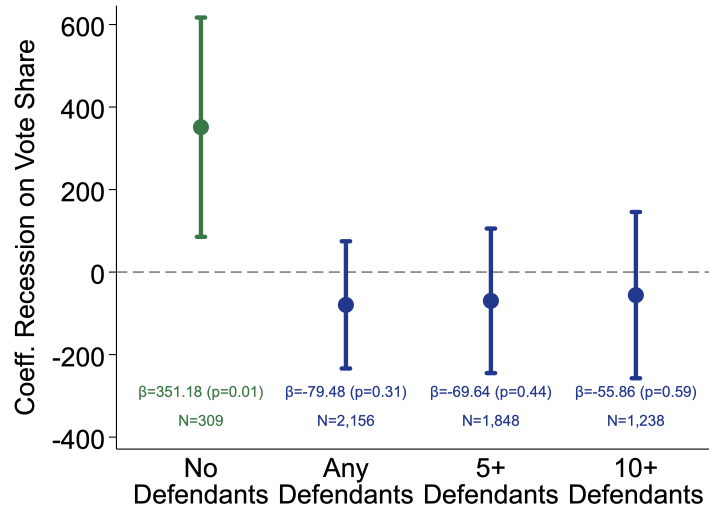
Mayors from parties with no *Lava Jato* defendants received an electoral premium: municipalities at the 75th percentile of recession exposure rewarded incumbents by 9.13 percentage points more ( $351.18 \times 0.026$ ) than those at the 25th percentile. This suggests that voters in areas more exposed to the *Lava Jato*-driven recession rewarded incumbents not associated with corruption. In contrast, for parties with at least one, more than five, or more than ten defendants, the point estimates are insignificant. Figure A7 further sorts parties with at least 50 re-election candidates by number of defendants and finds no clear monotonic relationship between the number of defendants and electoral punishment. One possible explanation is that party labels are a weak signal of federal politics at the local level, as discussed in Section 5. In summary, voters appear to reward mayors not associated with *Lava Jato*, but do not systematically punish those who are.

I then assess whether electoral punishment varies by party ideology. Given the mass protests

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<sup>7</sup>16 out of 26 parties had at least one defendant, and those 16 parties controlled 87% of mayorships that ran for re-election in 2016.

Figure 8: Effect of Recession Exposure on Vote Share by Number of *Lava Jato* Defendants



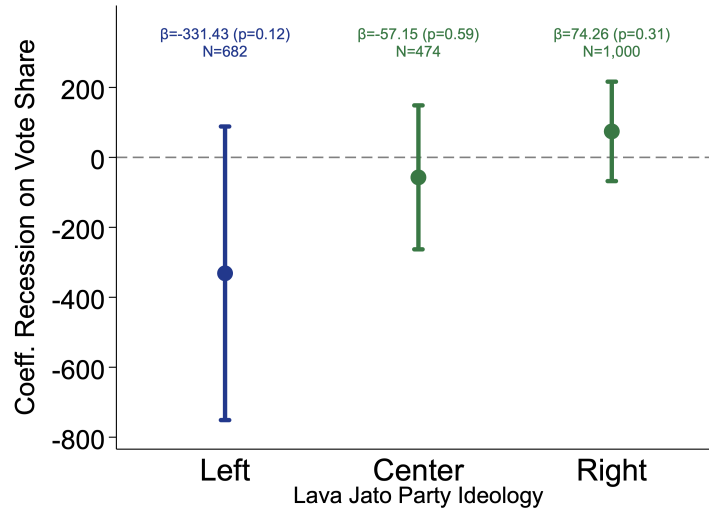
Notes: Each point displays the coefficient of recession exposure on the change in the incumbent mayor's vote share from Equation (9), estimated separately for four subsamples defined by the number of *Lava Jato* defendants in the incumbent's party from Oliveira and Teixeira (2022). Vertical bars represent 95% confidence intervals. *p*-values and sample sizes are reported below each coefficient.

against the left-wing federal government documented in Section 5, I hypothesize that voters would punish left-wing *Lava Jato* parties more than their center or right-wing counterparts. Following the ideology classification of Zucco and Power (2023),<sup>8</sup> Figure 9 presents the recession exposure coefficient using Equation (9) separately for incumbent mayors from left-wing, center, and right-wing *Lava Jato*-investigated parties.

For left-wing investigated parties, the coefficient is negative but not statistically significant ( $p = 0.12$ ): municipalities at the 75th percentile of recession exposure punished incumbents by 8.61 percentage points more ( $-331.43 \times 0.026$ ) than those at the 25th percentile. The coefficient for center parties is close to zero and insignificant, while for right-wing parties it is positive and insignificant. Overall, there is suggestive evidence that voters punished left-wing *Lava Jato* parties more than their counterparts, consistent with the broader anti-left sentiment that characterized the *Lava Jato* era and the concurrent mass protests.

<sup>8</sup>Ideology scores from Zucco and Power (2023) range from  $-1$  to  $1$ . Center parties are defined as those with scores between  $-0.2$  and  $0.2$ , left parties as those below  $-0.2$ , and right parties as those above  $0.2$ .

Figure 9: Effect of Recession Exposure on Vote Share by *Lava Jato* Party Ideology

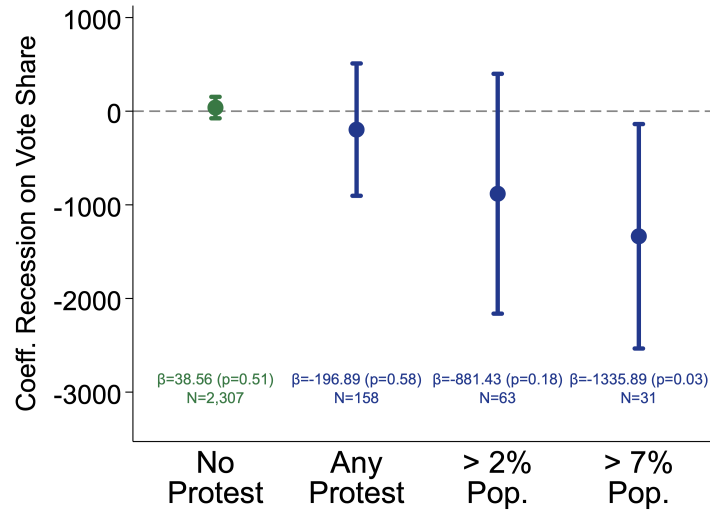


Notes: Each point displays the coefficient of recession exposure on the change in the incumbent mayor's vote share from Equation (9), estimated separately for incumbent mayors from left-wing, center, and right-wing parties investigated by *Lava Jato*. Party ideology follows Zucco and Power (2023). Vertical bars represent 95% confidence intervals. *p*-values and sample sizes are reported above each coefficient.

Amid rising media coverage of *Lava Jato*, mass protests took place between 2015 and 2017 against the PT federal government and in support of the investigations. I collect protest data from G1 (2016), a major Brazilian news outlet reporting both state police and organizer estimates. To avoid organizer bias, I use state police figures. The largest protest occurred on March 13, 2016, when between 3.6 and 6.9 million people gathered across 337 cities to demand the impeachment of President Rousseff and express support for *Lava Jato*.

Figure 10 reports the coefficient of recession exposure on vote share from Equation (9) across four different samples of protest activity on March 13, 2016: (i) municipalities with no protest ("No Protest"); (ii) municipalities with any protest ("Any Protest"); (iii) municipalities with over 2% of population participated, approximately the top 25% of distribution (">2% Pop."), and (iv) municipalities with over 7% of population participated, approximately the top 10% of distribution (">7% Pop."). If information shapes electoral accountability, economic voting should be more pronounced where protests were larger. Consistent with this hypothesis, the coefficient becomes increasingly negative as protest intensity rises. In municipalities without protest, the average voter does not punish the mayor for the recession. Where more than 2% of the population participated,

Figure 10: Effect of Recession Exposure on Vote Share by Mass Protests



*Notes:* Each point displays the coefficient of recession exposure on the change in the incumbent mayor's vote share from Equation (9), estimated separately for four subsamples defined by protest activity on March 13, 2016: (i) municipalities with no recorded protest; (ii) municipalities with any protest; (iii) municipalities where more than 2% of the population participated (approximately the top 25% of the distribution); and (iv) municipalities where more than 7% of the population participated (approximately the top 10% of the distribution). Protest data are from [G1 \(2016\)](#), using state police attendance estimates. Vertical bars represent 95% confidence intervals. *p*-values and sample sizes are reported below each coefficient.

the coefficient becomes negative but remains insignificant. Where more than 7% attended, the coefficient becomes statistically significant: municipalities at the 75th percentile of recession exposure punished incumbents by 22.71 percentage points more ( $-1,335.89 \times 0.017$ ) than those at the 25th percentile. The large magnitude reflects that municipalities with high protest activity are those where anti-government sentiment was most concentrated.

While general information proxies such as cell-phone and computer penetration do not tell a clear story, corruption signaling through party affiliation and mass protest activity together point to a more convincing mechanism. Building on [Ashworth et al. \(2018\)](#) and [De Benedictis-Kessner and Warshaw \(2020\)](#), recession shocks appear to make voters more attentive to corruption-related information. Where the incumbent belonged to a left-wing party investigated by *Lava Jato*, or where protest activity was more pronounced, economic voting is stronger. These findings suggest that access to political and corruption information plays a central role in local electoral accountability, consistent with the broader literature on media and accountability ([Ferraz and Finan, 2008](#);

Snyder and Strömberg, 2010).

## 7 Conclusion

This paper examines whether local exposure to Brazil’s 2014–2016 recession shaped municipal electoral outcomes, using shift-share instruments (Bartik, 1991) to measure local exposure to *Lava Jato* and decomposed China trade shocks (Costa et al., 2016). I find that *Lava Jato*-driven recession exposure did not affect incumbent mayors’ vote shares, and that despite their significant impact on local labour markets, China trade shocks did not influence municipal elections. Neither shock differentially affected PT-affiliated mayors. Nevertheless, voters punished the president for the *Lava Jato* shock and Chinese import competition, and rewarded them for Chinese export demand. These results are consistent with voters exercising sophistication in attributing responsibility across levels of government.

Furthermore, amid rising *Lava Jato* investigations and media coverage, I explore whether information about corruption shaped local electoral accountability. Mayors from parties with no *Lava Jato* defendants received an electoral premium, while there is suggestive evidence that left-wing *Lava Jato* parties faced greater punishment than their center or right-wing counterparts. In municipalities where more than 7% of the population attended mass protests against corruption, voters in more recession-exposed areas punished incumbents significantly more than those in less-exposed areas. This is consistent with an information channel that amplifies economic voting.

This paper has two main shortcomings. First, the RAIS dataset used to construct the shift of recession exposure covers only formal employment, despite approximately 50% of Brazilian workers being employed informally. I mitigate this concern by controlling for informality shares and examining subsamples above and below the informality median, finding no evidence of heterogeneity. Second, the information mechanism remains suggestive rather than causal. Media penetration proxies using cell-phone and computer access are not statistically significant, party labels are a weak signal of federal politics at the local level since mayors frequently switch parties, and the protest data covers few municipalities.

A promising avenue for future research is studying the causes of the recent rise of populism in Brazil. [Barros and Santos Silva \(2025\)](#) found that the 2014–2016 recession contributed to the rise of far-right populist former President Jair Bolsonaro in 2018. Similarly, I find that the import supply shocks rewarded right-wing mayors in the 2016 municipal elections is consistent with [Autor et al. \(2020\)](#), who argued that Chinese import competition increased support for conservative candidates in the United States, and suggests that similar dynamics may have developed at the local level in Brazil. A key challenge for this proposal is measuring populism at the municipal level. One approach is to construct a populism score through textual analysis of mayoral manifestos, and estimate the effect of the [Bartik \(1991\)](#) shift-share and decomposed [Costa et al. \(2016\)](#) China trade shocks on the vote share of populist candidates in Brazilian municipal elections.

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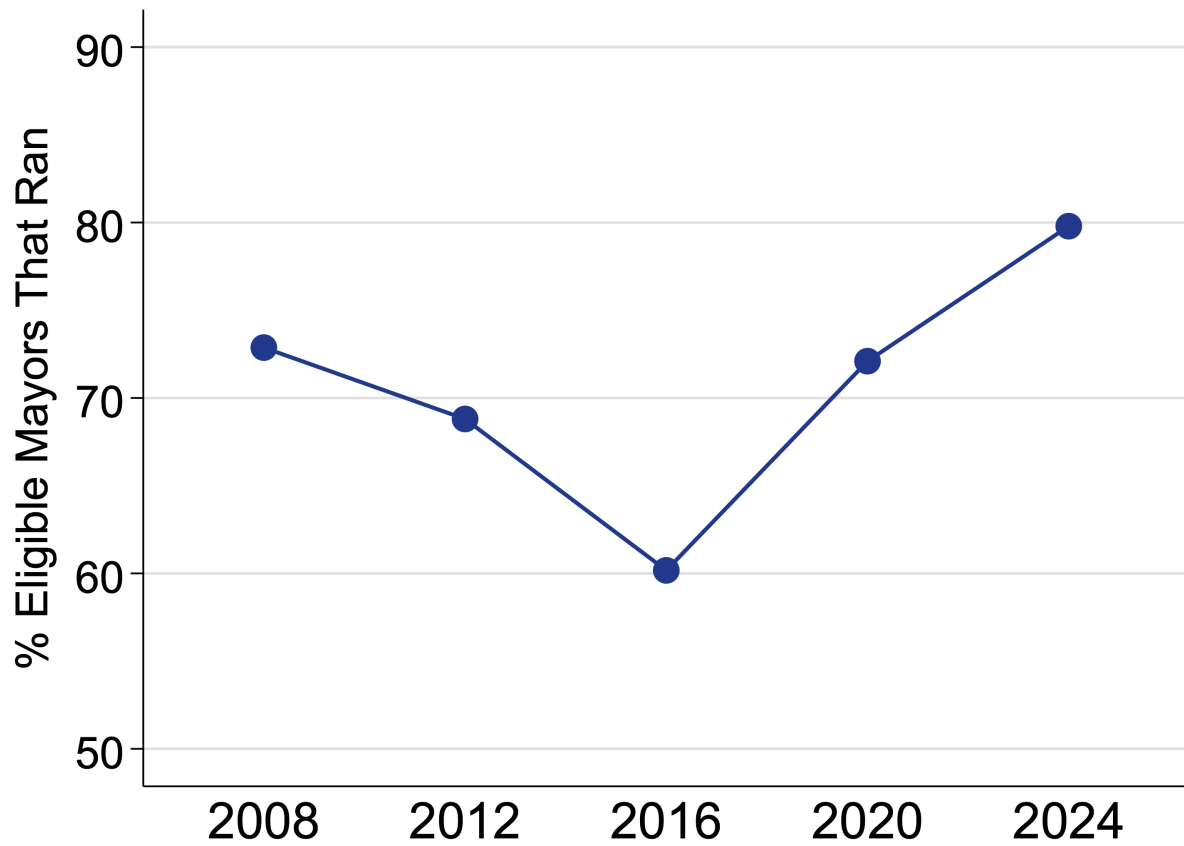
**Valarini, Elizangela and Markus Pohlmann**, “Organizational crime and corruption in Brazil a case study of the “Operation Carwash” court records,” *International Journal of Law, Crime and Justice*, 2019, 59, 100340.

**Wolfers, Justin**, “Are Voters Rational? Evidence from Gubernatorial Elections,” Working Paper 1730, Stanford Graduate School of Business 2002.

**Zucco, Cesar and Timothy J. Power**, “The Ideology of Brazilian Parties and Presidents: Coalitional Presidentialism Under Stress,” Working Paper 4470209, Social Science Research Network (SSRN) 7 2023.

## A Appendix

Figure A1: Percentage of Eligible Mayors that Ran for Re-Election

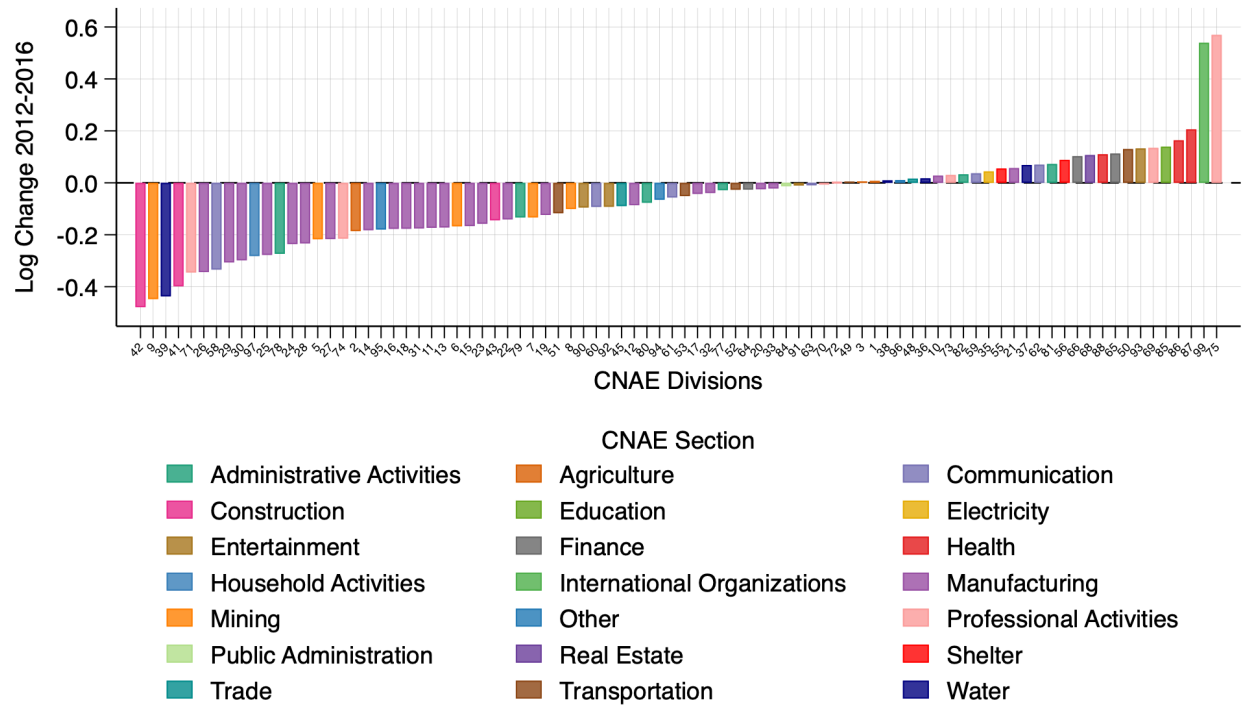


*Notes:* The figure plots the percentage of eligible incumbent mayors that ran for re-election across municipal election cycles. Eligible mayors are defined as those who won in the previous election and were not lame ducks. Data are from TSE.

Table A1: Economic Papers on the Effects of Economic Shocks on Political Outcomes in Brazil (sorted by year)

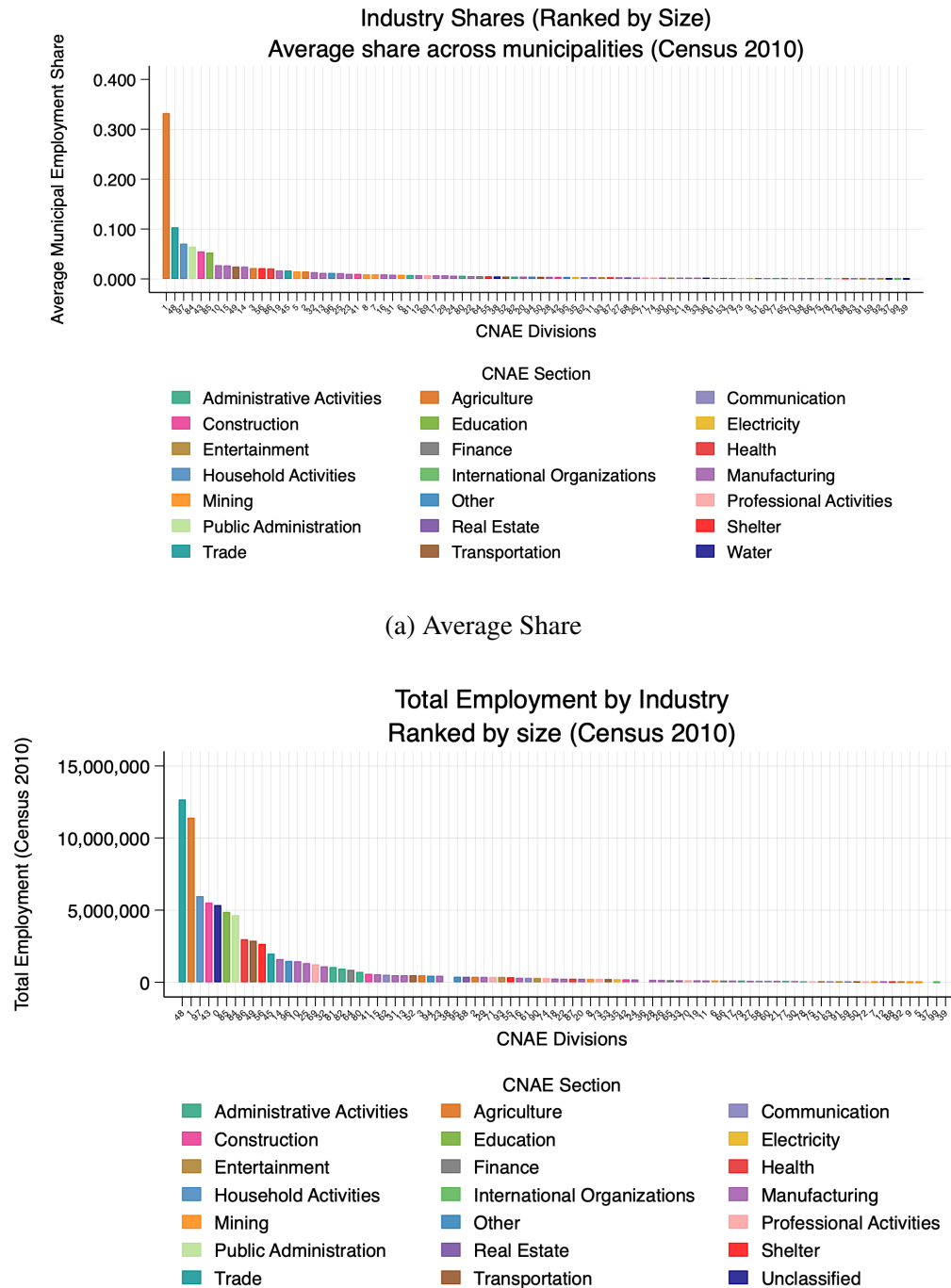
| Citation                                       | Method      | Field               | Question                                                                                                    | Duration                       | Location          | Obs.                                            | Main Results                                                                                                                                                                                                                                          | Mechanism                                                                                                                     |
|------------------------------------------------|-------------|---------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Costa et al. (2023)</a>            | Shift-Share | Trade & Religion    | Effects of economic downturns on the expansion of Pentecostal Evangelicalism                                | 1990–2010                      | Microregions      | 412 microregions and 161,124 religious entities | 1) $\downarrow$ SD tariffs $\Rightarrow$ $\uparrow$ 6.1 log points population share of Pentecostals; 2) $\downarrow$ SD tariffs $\Rightarrow$ $\uparrow$ 2% vote share of Pentecostal candidates; 3) Contributed to the election of Bolsonaro in 2018 | Social Insurance: 1) Presence of social assistance institutions offsets the effects; 2) Emotional struggle using suicide data |
| <a href="#">Ogeda et al. (2024)</a>            | Shift-share | Trade               | Trade liberalization affects the vote share of left-wing parties in presidential elections                  | 1994–2018                      | Microregions      | 484 microregions                                | $\uparrow$ SD of tariff reduction $\Rightarrow$ 4 p.p decrease in the vote share of left-wing presidential candidates                                                                                                                                 | Weakening of labour unions: 1) Decreased union presence; 2) Effects are driven by parties historically linked to unions       |
| <a href="#">Barros and Santos Silva (2025)</a> | Shift-Share | Labor and Gender    | Differential exposure to labor market shocks by gender contributed to Bolsonaro's rise (far-right populism) | 2014–17 crisis & 2018 election | Microregions      | 551 microregions                                | $\uparrow$ SD in male–female shock difference $\Rightarrow$ 0.6–0.9% increase in Bolsonaro's vote share                                                                                                                                               | Abortion surveys: men suffering large losses $\rightarrow$ support masculine/authoritarian narratives                         |
| <a href="#">Barros and Schmeißer (2025)</a>    | Dynamic DiD | Labor               | Job loss affects political entry and candidate quality                                                      | 2000–2020                      | Cities            | 994,214 treated and 994,214 controls            | 1) Job loss $\Rightarrow$ $\uparrow$ 12% joining a party (1–2 cycles) and 6–28% running for office; 2) Positive selection on valence (education, etc.)                                                                                                | $\downarrow$ Private-sector opportunity cost and $\uparrow$ Free time: UI eligibility RDD $\uparrow$ political entry          |
| <a href="#">Barros et al. (2025)</a>           | Panel FE    | Public & Behavioral | Exogenous economic conditions on electoral outcomes                                                         | 2010 & 2014 elections          | Brazilians abroad | $\approx$ 200,400 in 2010; 352,900 in 2014      | $\uparrow$ Host-country economic conditions $\Rightarrow$ $\uparrow$ Support for the incumbent                                                                                                                                                        | Rational and behavioral voter framework                                                                                       |

Figure A2: Industry Shifts 2016-2012 (Log)



*Notes:* Each bar displays the log change in national formal employment for a 2-digit CNAE industry between 2012 and 2016, computed from RAIS using a leave-one-out approach. Industries are sorted in ascending order by the magnitude of the shift. Colors denote CNAE sections as shown in the legend.

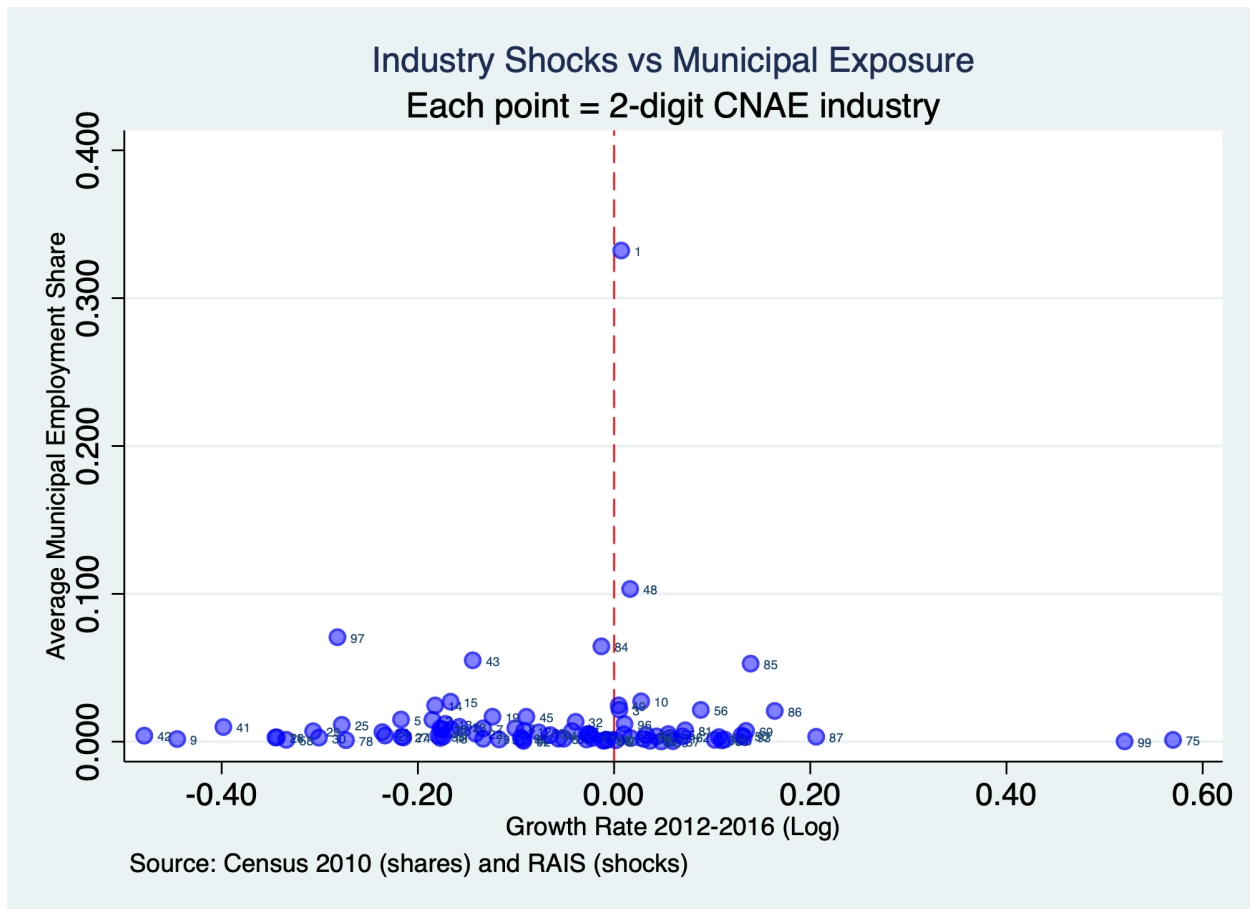
Figure A3: Industry Shares in 2010



*Notes:* Panel (a) displays the average municipal employment share for each 2-digit CNAE industry, computed from the 2010 Census. Panel (b) displays total employment by 2-digit CNAE industry from the 2010 Census. Industries are sorted in descending order by size. Colors denote CNAE sections.

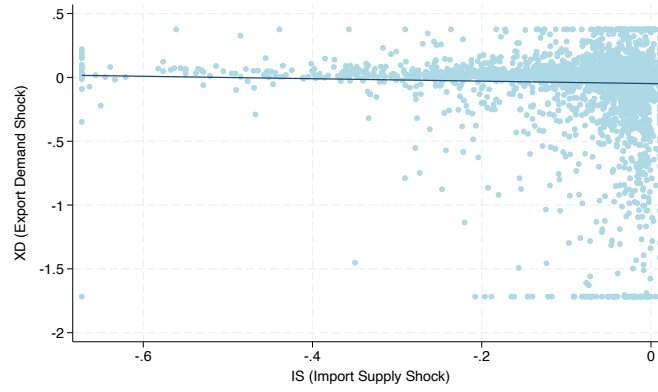


Figure A4: Average Shares vs. Shifts

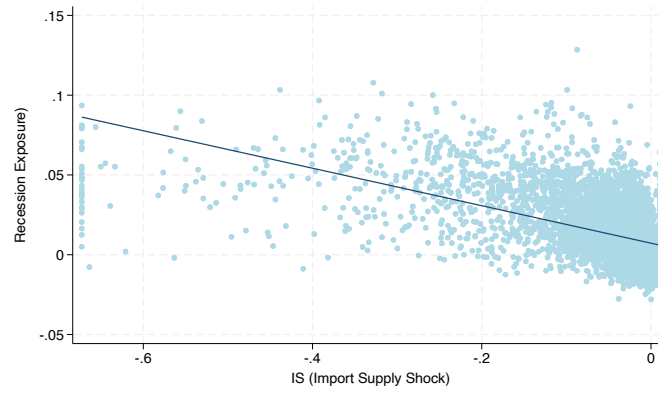


*Notes:* Each point represents a 2-digit CNAE industry. The horizontal axis shows the log change in national formal employment between 2012 and 2016 (the shift), computed from RAIS. The vertical axis shows the average municipal employment share in 2010 (the share), computed from the 2010 Census. Industry codes are labeled next to each point.

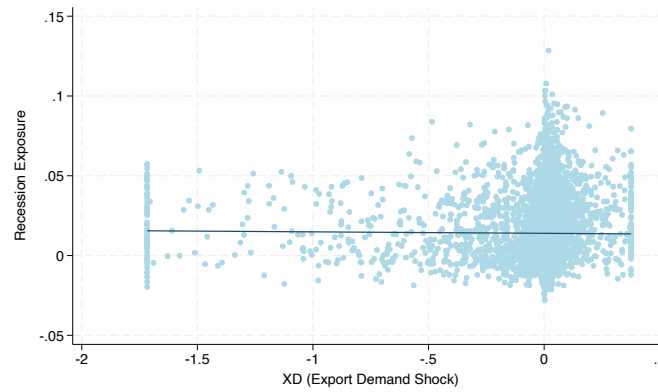
Figure A5: Correlation between National Shocks



(a) IS and XD



(b) Recession and IS



(c) Recession and XD

*Notes:* Panel (a) plots import supply (*IS*) against export demand (*XD*). Panel (b) plots recession exposure against import supply (*IS*). Panel (c) plots recession exposure against export demand (*XD*). Each observation is a municipality. Correlations are  $-0.02$  for *IS* and *XD*,  $-0.48$  for recession exposure and *IS*, and  $-0.01$  for recession exposure and *XD*. Trade variables are winsorized at the 1st and 99th percentiles.

Table A2: Robustness Checks: Incumbent Mayor Vote Share

|                                     | (1)<br>Dep. Var.<br>Coalition VS | (2)<br>Indep. Var.<br>2016–2013 | (3)<br>Indep. Var.<br>Level Diff. (in 10,000) |
|-------------------------------------|----------------------------------|---------------------------------|-----------------------------------------------|
| <i>Δ Incumbent Mayor Vote Share</i> |                                  |                                 |                                               |
| Recession Exposure                  | -21.37<br>(15.88)                | -15.60<br>(22.96)               | 0.35<br>(0.78)                                |
| <i>N</i>                            | 5,117                            | 2,465                           | 2,465                                         |
| Electoral Controls                  | ✓                                | ✓                               | ✓                                             |
| Demographic Controls                | ✓                                | ✓                               | ✓                                             |
| State Fixed Effects                 | ✓                                | ✓                               | ✓                                             |
| Mean(SD) of Recession               | 0.014(0.020)                     | 0.034(0.023)                    | -1.377(1.415)                                 |
| IQR of Recession                    | 0.027                            | 0.034                           | 2.034                                         |

*Notes:* Dependent variable is the change in incumbent mayor vote share (p.p.) between the 2016 and 2012 elections, unless otherwise stated. Column (1) uses the coalition vote share as the dependent variable, defined as the change in the vote share of the incumbent party whether running independently or as part of a coalition. Column (2) redefines the recession exposure shift-share using the 2016–2013 window as the independent variable. Column (3) uses the level difference in recession exposure scaled in 10,000 workers as the independent variable. All specifications include electoral controls, demographic controls, and state fixed effects. Column (3) is weighted by each municipality's share of the national workforce. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

Table A3: Effect of China Trade Shocks on Incumbent Party Vote Share

|                                                                  | (1)             | (2)              | (3)               | (4)              | (5)              |
|------------------------------------------------------------------|-----------------|------------------|-------------------|------------------|------------------|
| <i>Dep. Var.: <math>\Delta</math> Incumbent Party Vote Share</i> |                 |                  |                   |                  |                  |
| Import Supply                                                    | 1.49<br>(8.64)  | 2.46<br>(8.91)   | -11.15<br>(24.50) | -5.53<br>(21.78) | -4.19<br>(23.12) |
| Export Demand                                                    | -1.09<br>(2.49) | -0.92<br>(2.48)  | -1.63<br>(3.76)   | -3.75<br>(3.22)  | -3.68<br>(3.22)  |
| Recession Exposure                                               |                 | 58.50<br>(54.02) |                   |                  | 42.63<br>(66.04) |
| KP F-stat                                                        |                 |                  | 20.65             | 21.73            | 19.85            |
| <i>N</i>                                                         | 3,469           | 3,469            | 3,469             | 3,469            | 3,469            |
| Estimation Method                                                | OLS             | OLS              | IV                | IV               | IV               |
| Electoral Control                                                | ✓               | ✓                | ✓                 | ✓                | ✓                |
| Industry Controls                                                | ✓               | ✓                | ✓                 | ✓                | ✓                |
| Demographic Controls                                             | ✓               | ✓                | ✓                 | ✓                | ✓                |
| State Fixed Effects                                              | ✓               | ✓                |                   | ✓                | ✓                |
| IQR of Recession Exposure                                        | 0.028           | 0.028            | 0.028             | 0.028            | 0.028            |

*Notes:* Dependent variable is the change in the incumbent party's vote share (p.p.) between the 2016 and 2012 elections. Columns (1)–(2) report OLS estimates. Columns (3)–(5) report IV estimates instrumenting *IS* and *XD* with shift-share instruments based on China's bilateral trade with all countries excluding Brazil. All controls are as described in Table 1. All regressions are weighted by each municipality's share of the national workforce. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

Table A4: Effect of China Trade Shock on Incumbent Vote Share using [Autor et al. \(2020\)](#)'s Estimation Strategy

|                                                                | (1)               | (2)              | (3)               |
|----------------------------------------------------------------|-------------------|------------------|-------------------|
| <i>Panel A: <math>\Delta</math> Incumbent Mayor Vote Share</i> |                   |                  |                   |
| Import Supply                                                  | 3.982*<br>(2.128) | 2.537<br>(2.010) | 1.032<br>(2.211)  |
| <i>F</i> -statistic                                            |                   |                  | 46.68             |
| <i>N</i>                                                       | 2,439             | 2,436            | 2,439             |
| <i>Panel B: <math>\Delta</math> Incumbent Party Vote Share</i> |                   |                  |                   |
| Import Supply                                                  | 0.912<br>(1.766)  | 2.185<br>(1.751) | -1.322<br>(1.974) |
| <i>F</i> -statistic                                            |                   |                  | 62.18             |
| <i>N</i>                                                       | 3,433             | 3,429            | 3,433             |
| Estimation                                                     | OLS               | OLS              | 2SLS              |
| Census Controls                                                |                   | ✓                | ✓                 |
| Previous Vote Share                                            |                   | ✓                | ✓                 |
| State Fixed Effects                                            |                   | ✓                | ✓                 |

*Notes:* Dependent variable is the change in vote share (p.p.) between the 2016 and 2012 elections. Panel A uses the incumbent mayor's vote share and Panel B uses the incumbent party's vote share. The specification follows [Autor et al. \(2020\)](#). Column (3) instruments with a shift-share based on China's imports from Turkey, Mexico, Indonesia and Russia (EAGLES). Census controls include demographic and industry controls from the 2000 Census. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

Table A5: Effect of China Trade Shocks on Incumbent Mayor Vote Share by Informality

|                                                                  | (1)<br>Below Median | (2)<br>Above Median |
|------------------------------------------------------------------|---------------------|---------------------|
| <i>Dep. Var.: <math>\Delta</math> Incumbent Mayor Vote Share</i> |                     |                     |
| Import Supply                                                    | 1.02<br>(34.01)     | -2.95<br>(31.73)    |
| Export Demand                                                    | 0.26<br>(4.66)      | 2.05<br>(8.91)      |
| Recession Exposure                                               | -39.26<br>(117.82)  | -4.72<br>(54.74)    |
| KP F-stat                                                        | 11.99               | 5.85                |
| <i>N</i>                                                         | 1,233               | 1,232               |
| Electoral Control                                                | ✓                   | ✓                   |
| Industry Controls                                                | ✓                   | ✓                   |
| Demographic Controls                                             | ✓                   | ✓                   |
| State Fixed Effects                                              | ✓                   | ✓                   |
| IQR of Recession Exposure                                        | 0.033               | 0.014               |

*Notes:* Dependent variable is the change in the incumbent mayor's vote share (p.p.) between the 2016 and 2012 elections. The sample is split at the median of the informal employment share from the 2010 Census. IQR is interquartile range, which refers to the difference between the 75th and 25th percentiles of recession exposure within each subsample. All specifications include electoral controls, industry controls, demographic controls, and state fixed effects. All regressions are weighted by each municipality's share of the national workforce. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

Table A6: Effect of China Trade Shocks on Incumbent Party Vote Share by Sector Exposure

|                                              | (1)<br>Below Median | (2)<br>Above Median |
|----------------------------------------------|---------------------|---------------------|
| <i>Panel A: Agriculture &amp; Extractive</i> |                     |                     |
| Import Supply                                | -12.69<br>(34.09)   | 9.00<br>(53.01)     |
| Export Demand                                | -2.41<br>(4.32)     | 1.90<br>(12.65)     |
| KP F-stat                                    | 12.81               | 2.67                |
| N                                            | 1,232               | 1,232               |
| <i>Panel B: Manufacturing</i>                |                     |                     |
| Import Supply                                | 236.00<br>(158.60)  | -29.50<br>(36.61)   |
| Export Demand                                | -0.14<br>(3.77)     | -3.50<br>(5.97)     |
| KP F-stat                                    | 7.51                | 11.36               |
| N                                            | 1,232               | 1,231               |
| Estimation Method                            | IV                  | IV                  |
| Full Controls                                | ✓                   | ✓                   |
| State Fixed Effects                          | ✓                   | ✓                   |

*Notes:* Dependent variable is the change in the incumbent party's vote share (p.p.) between the 2016 and 2012 elections. The sample is split at the median of each sector's employment share from the 2010 Census. Panel A splits by combined agriculture and extractive employment share. Panel B splits by manufacturing employment share. All specifications include full controls and state fixed effects. All regressions are weighted by each municipality's share of the national workforce. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

Table A7: Effect of Shocks on PT Switching

|                                       | (1)                 | (2)             |
|---------------------------------------|---------------------|-----------------|
|                                       | PT Candidate Switch | PT Mayor Switch |
| <i>Dep. Var.: Switch Away from PT</i> |                     |                 |
| Import Supply (IS)                    | −1.05<br>(0.80)     | −1.99<br>(1.55) |
| Export Demand (XD)                    | −0.38<br>(0.58)     | 0.01<br>(0.39)  |
| Recession Exposure                    | −1.96<br>(2.85)     | −5.93<br>(4.01) |
| KP F-stat                             | 2.60                | 1.65            |
| <i>N</i>                              | 252                 | 141             |
| Estimation                            | IV                  | IV              |
| Industry Controls                     | ✓                   | ✓               |
| Demographic Controls                  | ✓                   | ✓               |
| State Fixed Effects                   | ✓                   | ✓               |

*Notes:* Dependent variable is an indicator for switching away from PT between the 2012 and 2016 municipal elections. Column (1) restricts to municipalities where the same candidate ran in both the 2012 and 2016 elections regardless of party affiliation, capturing whether a candidate who previously ran under PT switched parties by 2016. Column (2) restricts to municipalities where the PT candidate won in 2012 and stood for re-election in 2016, capturing whether an incumbent PT mayor switched party affiliation for their re-election. All columns report IV estimates instrumenting *IS* and *XD*. Industry and demographic controls are described in Table 1. All regressions are weighted by each municipality's share of the national workforce. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .



Table A8: Effect of Shocks on Incumbent Mayor Vote Share by Ideology

|                                                | (1)                 | (2)                 | (3)                | (4)                | (5)                | (6)                |
|------------------------------------------------|---------------------|---------------------|--------------------|--------------------|--------------------|--------------------|
|                                                | Left                |                     | Center             |                    | Right              |                    |
| <i>Dep. Var.: Δ Incumbent Mayor Vote Share</i> |                     |                     |                    |                    |                    |                    |
| Import Supply (IS)                             | −81.06<br>(60.10)   | −80.70<br>(59.88)   | −52.04<br>(30.28)  | −36.40<br>(30.46)  | 62.18**<br>(27.95) | 61.83**<br>(27.72) |
| Export Demand (XD)                             | 0.73<br>(4.87)      | 0.60<br>(4.89)      | −6.12<br>(5.65)    | −10.05<br>(7.42)   | −7.92<br>(6.76)    | −4.47<br>(6.01)    |
| Recession Exposure                             | −254.08<br>(175.30) | −254.52<br>(175.03) | −52.26<br>(107.80) | −22.35<br>(124.13) | 97.69<br>(65.55)   | 91.32<br>(71.30)   |
| KP F-stat                                      | 5.16                | 5.20                | 9.86               | 7.30               | 9.89               | 9.62               |
| N                                              | 792                 | 788                 | 486                | 444                | 1,185              | 1,109              |
| Ideology Classification                        | Comprehensive       | Restricted          | Comprehensive      | Restricted         | Comprehensive      | Restricted         |
| Electoral Controls                             | ✓                   | ✓                   | ✓                  | ✓                  | ✓                  | ✓                  |
| Industry Controls                              | ✓                   | ✓                   | ✓                  | ✓                  | ✓                  | ✓                  |
| Demographic Controls                           | ✓                   | ✓                   | ✓                  | ✓                  | ✓                  | ✓                  |
| State Fixed Effects                            | ✓                   | ✓                   | ✓                  | ✓                  | ✓                  | ✓                  |

*Notes:* Dependent variable is the change in the incumbent mayor's vote share (p.p.) between the 2016 and 2012 elections. All columns report IV estimates instrumenting *IS* and *XD* with shift-share instruments based on China's bilateral trade with all countries excluding Brazil. Comprehensive and restricted classification is explained in footnote 5. Electoral, industry, and demographic controls are described in Table 1. All regressions are weighted by each municipality's share of the national workforce. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

Table A9: Effect of Recession Exposure (2012–2016) on PT Presidential Vote Share

|                                                                        | (1)                  | (2)                  | (3)                 | (4)                 |
|------------------------------------------------------------------------|----------------------|----------------------|---------------------|---------------------|
| <i>Panel A: <math>\Delta</math> PT Vote Share, Round 1 (2018–2014)</i> |                      |                      |                     |                     |
| Recession Exposure <sub>12–16</sub>                                    | –100.01***<br>(5.93) | –131.97***<br>(6.03) | –55.10***<br>(6.27) | –32.39***<br>(4.71) |
| <i>N</i>                                                               | 5,565                | 5,565                | 5,565               | 5,565               |
| <i>Panel B: <math>\Delta</math> PT Vote Share, Round 2 (2018–2014)</i> |                      |                      |                     |                     |
| Recession Exposure <sub>12–16</sub>                                    | –145.31***<br>(5.11) | –141.66***<br>(5.46) | –44.43***<br>(5.55) | –35.30***<br>(5.23) |
| <i>N</i>                                                               | 5,565                | 5,565                | 5,565               | 5,565               |
| Electoral Control                                                      |                      | ✓                    | ✓                   | ✓                   |
| Demographic Controls                                                   |                      |                      | ✓                   | ✓                   |
| State Fixed Effects                                                    |                      |                      |                     | ✓                   |
| IQR of Recession Exposure                                              | 0.028                | 0.028                | 0.028               | 0.028               |

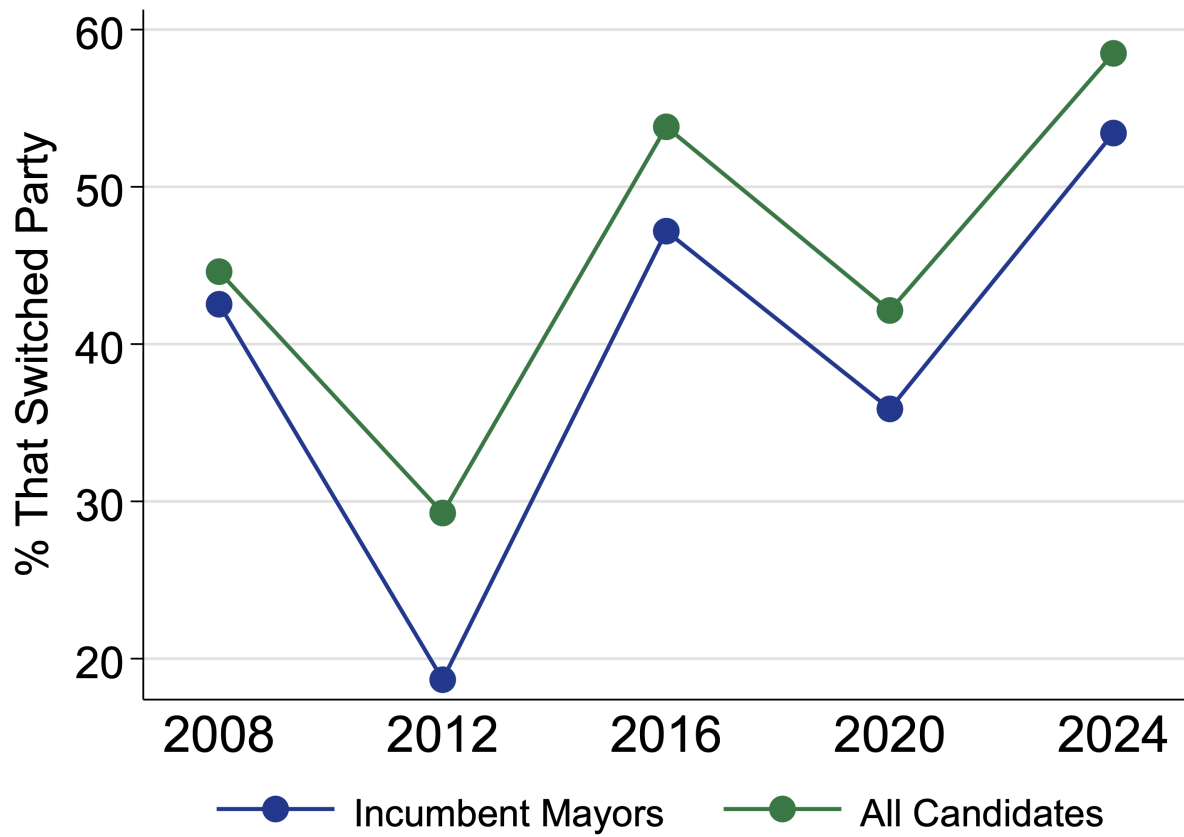
*Notes:* Dependent variable is the change in PT’s presidential vote share (p.p.) between the 2018 and 2014 elections. Panel A uses the first round and Panel B uses the second round. Recession exposure is the shift-share measure of local exposure to the 2012–2016 recession, using the same window as the mayoral election cycle. IQR is interquartile range which refers to the difference between the 75th and 25th percentiles of recession exposure. All controls are as described in Table 1. Robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

Table A10: Effect of China Trade Shocks (2012–2016) on PT Presidential Vote Share

|                                                                        | (1)               | (2)                 | (3)               | (4)               | (5)                 |
|------------------------------------------------------------------------|-------------------|---------------------|-------------------|-------------------|---------------------|
| <i>Panel A: <math>\Delta</math> PT Vote Share, Round 1 (2018–2014)</i> |                   |                     |                   |                   |                     |
| Import Supply <sub>16–12</sub>                                         | 0.37<br>(0.91)    | −0.51<br>(0.93)     | −3.26<br>(2.71)   | −0.49<br>(2.04)   | −0.99<br>(2.08)     |
| Export Demand <sub>16–12</sub>                                         | 0.98***<br>(0.30) | 0.93***<br>(0.30)   | 1.73***<br>(0.64) | 0.14<br>(0.49)    | 0.20<br>(0.49)      |
| Recession Exposure <sub>12–16</sub>                                    |                   | −20.78***<br>(5.58) |                   |                   | −21.92***<br>(6.04) |
| KP F-stat                                                              |                   |                     | 85.64             | 82.08             | 81.25               |
| <i>N</i>                                                               | 5,564             | 5,564               | 5,564             | 5,563             | 5,563               |
| <i>Panel B: <math>\Delta</math> PT Vote Share, Round 2 (2018–2014)</i> |                   |                     |                   |                   |                     |
| Import Supply <sub>16–12</sub>                                         | −0.28<br>(1.09)   | −1.44<br>(1.12)     | 1.78<br>(2.94)    | −1.42<br>(2.48)   | −2.02<br>(2.53)     |
| Export Demand <sub>16–12</sub>                                         | 1.10***<br>(0.36) | 1.03***<br>(0.35)   | 4.20***<br>(0.66) | 1.49***<br>(0.55) | 1.58***<br>(0.55)   |
| Recession Exposure <sub>12–16</sub>                                    |                   | −24.34***<br>(6.24) |                   |                   | −24.70***<br>(6.77) |
| KP F-stat                                                              |                   |                     | 85.50             | 82.08             | 81.25               |
| <i>N</i>                                                               | 5,564             | 5,564               | 5,564             | 5,563             | 5,563               |
| Estimation                                                             | OLS               | OLS                 | IV                | IV                | IV                  |
| Electoral Controls                                                     | ✓                 | ✓                   | ✓                 | ✓                 | ✓                   |
| Industry Controls                                                      | ✓                 | ✓                   | ✓                 | ✓                 | ✓                   |
| Demographic Controls                                                   | ✓                 | ✓                   | ✓                 | ✓                 | ✓                   |
| State Fixed Effects                                                    | ✓                 | ✓                   |                   | ✓                 | ✓                   |
| IQR of Import Supply                                                   | 0.059             | 0.059               | 0.059             | 0.059             | 0.059               |
| IQR of Export Demand                                                   | 0.054             | 0.054               | 0.054             | 0.054             | 0.054               |
| IQR of Recession Exposure                                              |                   | 0.028               |                   |                   | 0.028               |

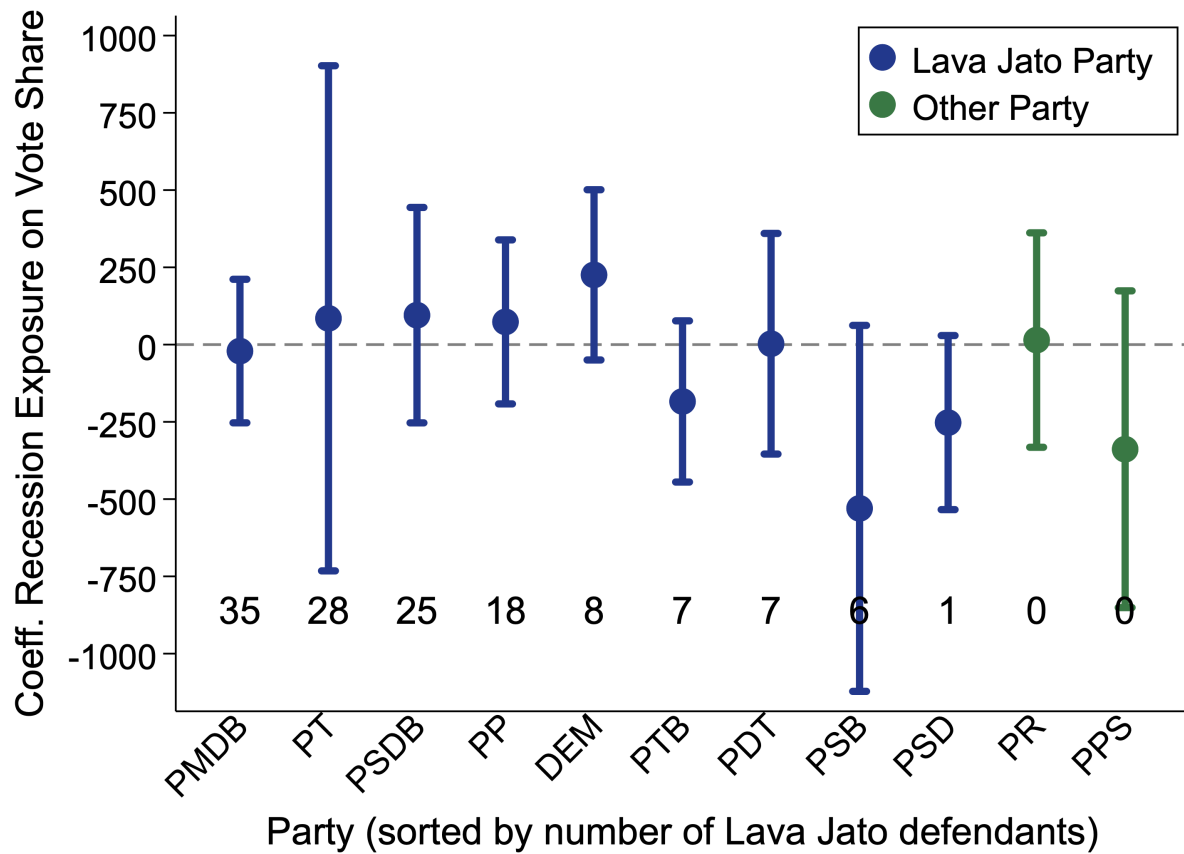
*Notes:* Dependent variable is the change in PT's presidential vote share (p.p.) between the 2018 and 2014 elections. Panel A uses the first round and Panel B uses the second round. Import supply and export demand are shift-share measures of local exposure to Chinese trade shocks over the 2012–2016 window, matching the mayoral election cycle. Recession exposure is the shift-share measure of local exposure to the 2012–2016 recession. Columns (1)–(2) report OLS estimates. Columns (3)–(5) report IV estimates instrumenting import supply and export demand with shift-share instruments based on China's bilateral trade with all countries excluding Brazil. Electoral controls include the 2010–2014 change in PT's presidential vote share. Industry and demographic controls are described in Table 1. Robust standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Figure A6: Percentage of Mayors and Candidates that Switched Parties



*Notes:* The figure plots the percentage of incumbent mayors (blue) and all candidates (green) that ran under a different party than in the previous election cycle, conditional on running in both elections. Data are from TSE.

Figure A7: Effect of Recession Exposure on Incumbent Mayor Vote Share



*Notes:* Each point displays the coefficient of recession exposure on the change in the incumbent mayor's vote share from Equation (9), estimated separately for each party. Parties are sorted in descending order by the number of *Lava Jato* defendants. Blue points denote parties with at least one *Lava Jato* defendant and green points denote parties with no defendants, based on [Oliveira and Teixeira \(2022\)](#). Only parties with at least 50 incumbent mayors running for re-election are included. All specifications include electoral controls, demographic controls, industry controls, and state fixed effects for parties with at least 100 observations. All regressions are weighted by each municipality's share of the national workforce. Vertical bars represent 95% confidence intervals. The number of defendants is reported below each coefficient.